



On-Orbit Servicing Networks in Cislunar Space: A Framework for Orbit Selection, Transfer Design, and Scheduling

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Abstract

This investigation introduces a comprehensive framework for designing and optimizing on-orbit servicing networks in cislunar space, integrating innovative approaches to orbit selection, transfer design, and initial scheduling. Within the context of this investigation, scheduling refers to the selection of the servicer-customer satellite pairings and the order in which each of the customer hubs are processed. A genetic algorithm serves as a basis for initial orbit selection, as well as search space reduction by adhering to beneficial criteria such as deployment costs, station-keeping requirements, and Lunar south pole coverage. The Genetic Algorithm is also leveraged to identify high-performing orbit families while supporting the extension to additional mission objectives. A phasing-informed transfer design pipeline is introduced, employing manifold arc trajectories and phasing penalties to estimate transfer costs efficiently for long-range rendezvous. This approach enhances the understanding of the underlying dynamics that is critical for trajectory design and schedule development. Two initial scheduling solutions are proposed: an Auction Algorithm with a low computational cost and a computationally intensive Genetic Algorithm. These strategies facilitate the creation of scalable servicing plans across diverse orbital configurations and a varying numbers of satellites. Trade space analysis further identifies optimal orbit configurations that align with mission requirements. This versatile and adaptable methodology offers critical insights into orbit selection, transfer trajectory design, and servicer scheduling options, delivering a robust foundation for designing servicing networks tailored to meet diverse mission objectives.

Keywords On-orbit servicing · Trajectory design · Three-body problem

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1 Introduction

On-Orbit Servicing (OOS), the process of maintaining, upgrading, refueling, or repositioning satellites while they remain operational in space, has gained significant momentum in recent years. The ability to service satellites in orbit offers strategic and economic advantages, enabling operators to extend mission lifetimes, enhance functionality, and reduce the overall cost of replacing assets. As space infrastructure becomes more critical to global communications, navigation, and research, the demand for servicing solutions has driven both government and commercial entities to develop and deploy satellite servicing capabilities.

Several companies and government organizations are leading the development in satellite servicing. Notable efforts include Northrop Grumman's Mission Extension Vehicle (MEV), that has successfully demonstrated the capability to extend the lifespans of geostationary satellites, and Astroscale's ELSA-d mission, that demonstrated debris removal and servicing technologies. NASA's OSAM-1 mission aimed to service satellites in Low Earth Orbit (LEO) but was discontinued due to budget constraints. Despite some setbacks, interest in satellite servicing remains strong, with most current initiatives targeting satellites in LEO or Geosynchronous Earth Orbit (GEO), where high asset values and dense operational networks justify the cost of these emerging technologies.

As exploration shifts beyond GEO, there is a growing need to establish servicing capabilities in the cislunar regime. NASA's Artemis program, along with plans for the Lunar Gateway, mark a renewed commitment to lunar exploration and, with it, the demand for dependable, nearby servicing options becomes paramount. Extending satellite servicing to cislunar space presents new challenges and opportunities, given the complex gravitational dynamics in the Earth-Moon system and the heightened logistical demands for deep-space missions. The ability to sustain a network of satellites in cislunar space would support a range of lunar and interplanetary missions, further enabling scientific research, navigation, surveillance, and communication networks to evolve in the Moon's vicinity.

In response to these challenges, this investigation introduces a framework for designing satellite servicing networks tailored for the cislunar environment. In this proposed architecture, a fleet of hubs, or larger spacecraft, act as dynamic "home bases", supporting smaller, agile servicer spacecraft. Each servicer satellite departs from its designated hub, travels to nearby customer satellites, performs the required servicing tasks, and returns to the hub to prepare for subsequent servicing missions. This formulation ensures that each servicer operates with minimal downtime, allowing a reliable, adaptable servicing network across cislunar space. There are clearly modifications that may improve performance, but the priority in this analysis is developing a basic, functional architecture as a baseline.

The design of such a servicing network draws parallels to the well known Traveling Salesman Problem (TSP), where the goal is determining the shortest possible route that allows a "salesman" to visit each "city" exactly once before returning to the starting point [1]. Extension to the Dynamic Traveling Salesman Problem (DTSP) [2] provides additional schemes for generating optimal solutions and evaluating the upper and lower bounds [3, 4]. And, when applied to the Orbit Traveling Salesman

Problem (OTSP), multi-step methods for decomposing the OTSP for use in LEO and GEO [5, 6] are available in the literature, but application to the lunar regime presents a unique set of challenges. Within the context of cislunar servicing, the TSP translates to determining an optimal sequence of satellite servicing stops that minimizes the propellant required (ΔV) or maximizes efficiency. The dynamical nature of orbits in the Earth-Moon system renders the problem a particularly complex variant of the TSP, denoted the Dynamic Traveling Salesman Problem (DTSP), where the distances between nodes (or servicing locations) vary over time.

Several approaches are known to provide solutions to the TSP and its dynamic variations, and these approaches are potentially applicable to satellite servicing networks. Designing tours for single servicer networks is similar to the concept of designing tours for debris mitigation [7] or asteroid visitation [8], but an extension to multiple servicers frequently requires the use of heuristic approaches. Common solution techniques include Ant Colony Optimization, that simulates the foraging behavior of ants to locate optimal paths; Genetic Algorithms, that evolve solutions over generations to identify high-fitness servicing schedules; and tree search algorithms, that evaluate branching paths to build an efficient solution space. By applying these methods to the scheduling of cislunar servicing missions, this framework achieves optimized sequences of servicing operations across a variety of orbital configurations. In this investigation, the term "scheduling" refers to the process of assigning customer spacecraft to corresponding servicer craft and the order in which each of the customer craft are processed. Related two-phase frameworks and campaign-level optimizations [9, 10] motivate the 2-phase decomposition in the GA but differ in two ways for cislunar OSAM: (1) dynamics must capture CR3BP/ephemeris effects that reshape timing and feasible manifolds; and (2) feasible orbits and transfers are constructed on demand, not pulled from a catalog. This phasing-informed pipeline generates heat-maps from manifold arcs and selects seeds consistent with rendezvous timing before scheduling, then converges for position continuity.

Scheduling in this domain departs from classical assignment or Traveling Salesperson formulations because feasible edge costs (ΔV and time) are constructed, not tabulated, and they depend on epoch, phasing, and the selected transfer family under CR3BP/ephemeris dynamics. As a result, the cost graph is dynamic, asymmetric, and time-dependent; feasibility and objective are jointly tied to boundary-value problems rather than to static distances. In this setting there are no closed-form solutions or standard convex relaxations that capture the coupling between dynamics and schedule. To make progress, two independent scheduling strategies on the same dynamics-consistent cost model are evaluated: (i) an auction-based method that allocates customers to servicers using phasing-aware edge evaluations, and (ii) a genetic-algorithm method that directly searches service orderings and rendezvous timing. These are alternative schedulers and together they provide a first, end-to-end sense of typical costs for OSAM networks rather than a single point design.

Beyond scheduling, the architecture for a servicing network depends heavily on orbit selection and transfer design. As most development activities involving IOS networks focuses on LEO and GEO, design strategies for servicer placement revolves around placing servicers near the most populated orbits. As such, Rouso et al. [11] propose a network of servicer satellites, equally spaced in phase, in a Sun-

synchronous orbit to reduce the inclination change when transferring to customer satellites. Hudson and Kolosa [12] and Xu et al. [13] design missions for satellite servicing in GEO; Braun et al. [14], and Yang et al. [15] provide debris removal mission scenarios in LEO considering a single servicing satellite de-orbiting multiple debris objects over the course of the mission. Across all LEO/GEO servicing/debris removal scenarios, the optimal configuration for the initial servicer satellite orbit is one where the inclination and right ascension of the ascending node are similar to that of the target objects, minimizing the total ΔV required for orbital transfers. Therefore, in IOS network design, selecting appropriate orbits for servicing hubs and customers is crucial for optimizing accessibility and minimizing transfer costs. For operations in cislunar space, this involves leveraging the Circular Restricted Three-Body Problem (CR3BP), a lower-fidelity model for the gravitational interactions between the Earth, Moon, and spacecraft to reveal potential transfer paths. By employing the CR3BP, invariant manifolds associated with unstable periodic orbits are frequently leveraged, enabling low-energy transfers between servicing hubs and customer satellites. These manifold trajectories offer a feasible approach to transfer design, providing preliminary insights into potential trajectories that are later validated and refined in higher-fidelity ephemeris models.

This work delivers a framework for designing satellite servicing networks in the cislunar region, encompassing hub and servicer orbit selection, optimized servicing schedules based on TSP principles, and efficient transfer design using CR3BP dynamics. By addressing the unique demands of cislunar space, the aim is advancing the feasibility of robust servicing networks to support future lunar and deep-space operations. The roadmap for this paper is as follows: Section 2 presents the dynamical models (CR3BP and N-body ephemeris) that underpin the framework. Section 3 details orbit selection criteria and the orbit families considered. Section 4 develops a phasing-informed transfer design pipeline and its heat-map interpolation strategy. Section 5 introduces two scheduling methods, an auction-based approach and a GA tour optimizer, and demonstrates each with illustrative, small-scale examples. Section 6 ('Results and Discussion') then presents end-to-end network outcomes, including ablation studies on network variants and validation in an N-body ephemeris model. Section 7 summarizes findings and outlines future work.

2 Dynamics Models

2.1 Circular Restricted Three-Body Problem

The Circular Restricted Three-Body Problem (CR3BP) is a foundational conceptual model in celestial mechanics that describes the motion of a small object (e.g., a satellite) influenced by the gravitational pull of two much larger bodies (such as the Earth and the Moon), that are assumed to orbit their mutual center of mass. In this model, the two spherically symmetric gravitational bodies, denoted the Earth and the Moon, constitute the primary system. They are assumed to move in circular orbits around their shared barycenter where the subscript "1" arbitrarily denotes the larger primary. The relative sizes of the primaries are represented by the mass ratio $\mu = \frac{m_2}{m_1 + m_2}$. The

spacecraft motion is described relative to a coordinate frame, $[\hat{x}, \hat{y}, \hat{z}]$, that rotates with the motion of the Earth and Moon. The nondimensional position and velocity vectors of the spacecraft relative to the barycenter form a six-dimensional state vector, $\bar{r} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T$. The scalar CR3BP equations of motion in the rotating frame are written

$$\frac{\delta U}{\delta x} = \ddot{x} - 2\dot{y}, \quad \frac{\delta U}{\delta y} = \ddot{y} + 2\dot{x}, \quad \frac{\delta U}{\delta z} = \ddot{z} \quad (1)$$

where the pseudo-potential U is defined as $U = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{d} + \frac{\mu}{r}$. The distances between the spacecraft and the first and second primary bodies are defined as $d = \sqrt{(x + \mu)^2 + y^2 + z^2}$ and $r = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}$, respectively.

The equations of motion in the CR3BP provide a single conserved scalar quantity, labeled the Jacobi Constant (JC). The JC is associated with the orbital energy as viewed in the rotating reference frame and remains constant along any ballistic trajectory. It consists of a pseudo-potential term combined with the square of the velocity in the rotating frame, evaluated as $JC = 2U - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. The Jacobi Constant value offers insight into the energy level for the various orbits, providing a foundation for comparing transfer costs and bounded regions in space.

2.2 Ephemeris Model

The N -body ephemeris serves as a dynamic model to simulate the gravitational interactions between multiple celestial bodies, each as a point mass. In this model, each celestial body is represented by its mass and position, without considering physical dimensions or rotational effects. The gravitational force between each pair of bodies is governed by Newton's law of universal gravitation. For accurate results for the positions and velocities of these bodies, the ephemerides provide precise, time-tagged positional data based on observations and complex models for the solar system dynamics. The instantaneous locations for the perturbing bodies relative to the central primary are accessed using NAIF SPICE ephemeris data, specifically DE421 from NASA Jet Propulsion Lab (JPL).

The N -body ephemeris model provides a higher-fidelity environment over simpler models, such as the two-body problem or the CR3BP, because it incorporates the full gravitational influence of multiple bodies in the solar system. In the ephemeris model, the spacecraft P_i is modeled in an inertial frame $\mathcal{I} = [\hat{X}, \hat{Y}, \hat{Z}]$ centered at one of the primaries P_q . The equations of motion in the N -body model are expressed

$$\ddot{\bar{r}}_{qi} = -G \frac{M_i + M_q}{r_{qi}^3} \bar{r}_{qi} + G \sum_{j=0, j \neq i, q}^N M_j \left(\frac{\bar{r}_{ij}}{r_{ij}^3} - \frac{\bar{r}_{qj}}{r_{qj}^3} \right) \quad (2)$$

where G is the gravitational constant and the vectors $\bar{r}_{ij}$ denote the vector from i to j . In this work, unless noted, the spacecraft is propagated in the Moon-centered J2000 frame incorporating the gravitational effects of the Earth, Moon, and Sun. The Moon-

centered approach is selected due to the close vicinity of many orbits of interest to the Moon [16]. In scenarios where the three body orbits spend more time in proximity to the Earth, as is the case with many resonant orbits, the dynamics are propagated and converged in the Earth-centered J2000 frame.

2.3 Periodic Orbits and Properties

Periodic orbits in the CR3BP are closed trajectories where P_3 , that is a spacecraft, returns to its initial position and velocity after a fixed period, T , repeating its motion indefinitely in the rotating reference frame. These orbits can offer stable or nearly-stable paths for spacecraft in multi-body gravitational environments. Individual periodic orbits belong to larger sets of orbit families. In families of periodic orbits, characteristics such as the period, Jacobi Constant, and state vectors evolve over smooth continuous curves. Families are constructed by employing a differential corrections algorithm to converge a single orbit for periodicity; then, varying a parameter (such as the JC) a continuation scheme aids to converge additional family members. One sample family of periodic orbits in the CR3BP is the planar Lyapunov family. While Lyapunov orbits exist in the vicinity of the Lagrange points L_1 , L_2 , and L_3 , a subset of the L_1 family is plotted in Fig. 1a. For the Lyapunov family, a single shooter corrections scheme targets a specified Jacobi Constant value, while satisfying the requirement for periodicity, delivering 20 sample members from the family.

The State Transition Matrix (STM), denoted $\Phi(t_1, t_0)$, is a crucial tool for any corrections process in orbital dynamics. It represents a linear mapping and provides a linear approximation for variations in the state elements as they evolve along a trajectory in response to a perturbation in the initial state. For a periodic orbit, the STM corresponding to precisely one orbital period is denoted the monodromy matrix

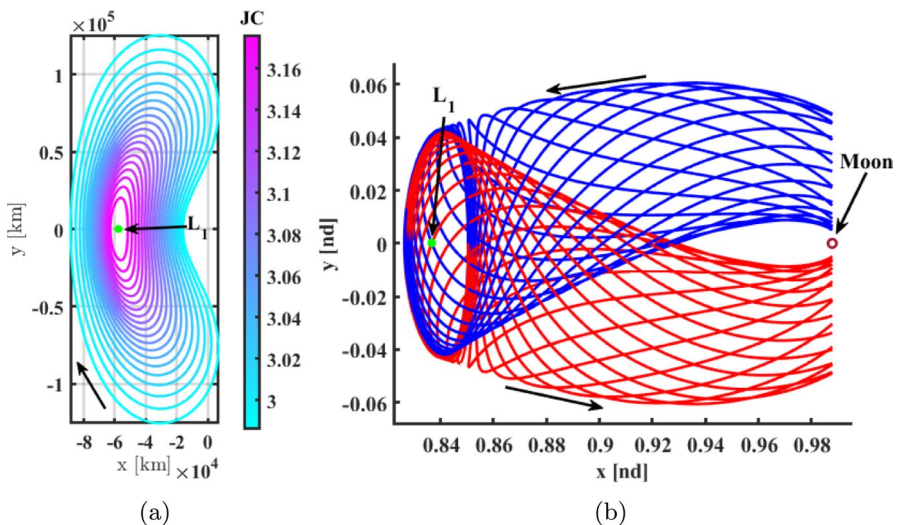


Fig. 1 **a** L_1 Lyapunov Orbits in the CR3BP and **b** The half manifolds corresponding to one L_1 Lyapunov orbit as viewed in configuration space

$(\Phi(T, 0))$, that offers insight into the linear stability of the orbit. Due to the Hamiltonian nature of the CR3BP, the six eigenvalues of the monodromy matrix always appear in reciprocal pairs. Notably, one pair of eigenvalues remains equal to unity, reflecting the inherent periodicity of the orbit. Any eigenvalue with a magnitude greater than one indicates linear instability in the associated solution [17]. An eigenvalue greater than one not only signals linear instability, but also identifies members in periodic orbit families with stable and unstable manifolds. These invariant manifolds represent higher-dimensional surfaces that asymptotically approach or depart the underlying orbit. By perturbing the states near the periodic orbit in the stable and unstable subspaces, the resulting trajectories lie along the corresponding stable and unstable manifold surfaces. For instance, Fig. 1b illustrates trajectories along manifolds projected into configuration space for one of the Lyapunov orbits. Manifold trajectories evolve in both positive and negative directions. In the figure, the blue trajectories along the stable manifold represent arcs that asymptotically approach the orbit from the lunar vicinity, while the red trajectories along the unstable manifold depict arcs that depart from the orbit toward the Moon. The green star marks the location of the L_1 Lagrange point.

Another useful parameter for evaluating the stability of an orbit is the time constant τ . The time constant is an estimate for the length of time required for a small perturbation along the state of the underlying orbit to grow by a factor of e . The time constant is written as,

$$\tau = \frac{1}{|Re(\ln(\lambda_{max}(\Phi(T, 0))))|} \quad (3)$$

where λ_{max} represents the largest eigenvalue from the monodromy matrix, $\Phi(T, 0)$ [18]. The time constant is a useful tool for evaluating the stability of orbits that offer longer periods, such as some resonant orbits.

In the subsequent sections, background into the strategies for orbit selection are reviewed; then, transfer design and initial scheduling are introduced. The next three sections present strategies to select servicing/customer orbits that offer the greatest utility, the process for designing phasing-informed transfers throughout servicing networks, and two methods for determining initial schedules that minimize total propulsive requirements. While a thorough derivation of the GA is introduced in the scheduling section, the GA also serves as an evaluation tool during the orbit selection process.

3 Orbit Selection

The proper orbits for satellite servicing are characterized by the selection of servicer and customer orbit pairings that supply the most efficient transfer and rendezvous scenarios. While orbit selection in LEO/GEO is straightforward, that is, the relative RAAN and relative inclination of the orbits should be minimized, the chaotic regime in cislunar space poses additional challenges. For this preliminary analysis, the search space for placement of servicer and customer orbits is limited to families

of periodic orbits that are known to exist in the CR3BP, already an infinite number, and the notion of using orbital elements centered on the Moon does not necessarily offer reduced transfer costs. However, there are some parallels to selecting similar characteristics, such as selecting orbits at similar energy levels (JC). To characterize the serviceability of a network, some criteria are summarized in Table 1 [19]. This list is a convenient approach to initially explore the selection of customer and servicer orbits. Realistically, customer orbits might be selected once the servicer orbits are already known or the opposite could occur. But for this analysis, both categories of orbits are selected in tandem to maximize the serviceability of the networks. The list is not all-encompassing and the criteria in Table 1 are not in any particular order. Near-resonant orbits that exploit symmetry allow repeatable geometries to support servicing activities over time. Orbital characteristics such as linear instability and similar energy levels between hub and customer orbits reduce the ΔV requirements. Phasing is one of the more challenging aspects of the problem, i.e., a rendezvous requirement for transfers to customer orbits and returns; thus, characteristics that aid such analysis are a priority.

In Waldecker and Howell [19], the number of servicers and customers are demonstrated to contribute to total network costs. As the number of servicer spacecraft are increased, ΔV per servicer spacecraft generally reduces, while increasing the number of customer satellites does not reduce the expected ΔV per customer satellite. While similarities in position space and JC are observed to reduce ΔV requirements, additional customers (even belonging to the same family) present significant phasing challenges. Imperfect initial conditions among the customer orbits, and varying underlying periods, leads to increased phasing requirements that outweigh the advantages of similarities in position.

The list of criteria offers some initial insight into the selection of orbits in a servicing framework, but it is not all-encompassing and is applied heuristically. Servicer scheduling and evaluation are accomplished using a Genetic Algorithm (GA). In addition to selecting the optimal servicing schedule, the GA also aids in trade space analysis for initial servicer and customer orbits that best satisfy mission constraints. To validate the proposed criteria for orbit selection, the fitness evaluation process in the GA consists of three components to achieve the goal of selecting orbits that reduce total network ΔV_T while also meeting mission requirements. In addition to incorporating the total network ΔV_T requirements, a penalty is applied for each additional servicer assigned to a tour (n_s) and a customer benefit is incorporated to

Table 1 Initial Design Criteria and Corresponding Advantages in Orbit Selection

Orbital criteria	Advantage
Linearly unstable	Use of hyperbolic manifolds in transfer design
Similar JCs	Lower ΔV costs for bridging energy gaps
Relatively "short" periods	More opportunities for favorable phasing
Near-resonant periods	Ease of schedule design
Exploits symmetry	Informs transfer design choices
Near period-multiplying bifurcations	Offers potential servicer locations and phasing orbits

encourage the inclusion of the best associated customers. While customer benefit is largely mission specific, three values are included to represent low deployment cost, low upkeep cost, and Lunar south pole coverage for this particular assessment. Therefore, the fitness score for a chromosome is evaluated as,

$$score = \frac{1}{\Delta V_T} - \gamma(n_s - 1) + \sum(\gamma_{JC}\beta_{JC}, \gamma_{SP}\beta_{SP}, \gamma_{SK}\beta_{SK}) \quad (4)$$

Note that a servicer penalty (the second term) is included to discourage the use of additional servicers in the network. While many optional servicers may be input to the algorithm, the network that uses the fewest servicers, with reasonable ΔV impact, is preferred. The scaling factor γ is included to model the cost of new servicer deployment. This value relates to the cost to deploy a new servicing hub or the maximum ΔV allowed using a single hub before additional servicers should be added. The third term, comprised of weighting values (γ_i) and benefit values (β_i), corresponds to the overall "benefit" of each customer orbit added to the network. More details on each of the variables follows.

The customer benefit is comprised of the satellite deployment benefit, (β_{JC}), stationkeeping benefit, (β_{SK}), and its south pole coverage benefit, (β_{SP}). Although ballistic lunar transfers are a low-cost option for satellite deployment in the lunar vicinity, the deployment costs are evaluated here as an energy differential between the orbit of interest and the L_1 and L_2 Lagrange points. A NASA plan to deploy satellites to halo orbits in the vicinity of the L_1 and L_2 Lagrange points is a part of a broader initiative known as the METrO campaign [20]. This campaign focuses on establishing a network of spacecraft to enhance communications and navigation in the vicinity of the Moon, particularly in support of the Artemis program and future lunar exploration missions. Therefore, the primary deployment cost after METrO is established involves transferring from either L_1 or L_2 halo orbits to the target orbit, a cost that is directly correlated to a change in the energy level. Since the specific halo orbit targets are yet to be determined, the energy levels associated with the Earth-Moon Lagrange points are used as an alternative. Therefore, the deployment benefit value is evaluated as,

$$\beta_{JC} = \frac{1}{1 + \min(|JC - JC_{L_1}|, |JC - JC_{L_2}|)} \quad (5)$$

As the linear instability associated with an orbit increases, the estimated costs to conduct stationkeeping maneuvers also increases. While it is difficult to precisely predict the stationkeeping costs from the stability of an orbit, they are directly correlated, and can inversely represent the stationkeeping benefit. First, the time constant is introduced, an estimate of the number of revolutions along an orbit for a perturbation to grow by a factor of e , i.e.,

$$\tau = \frac{1}{|Re(\ln(\lambda_{max}(\Phi(T, 0))))|} \quad (6)$$

where λ_{max} represents the largest eigenvalue from the monodromy matrix, i.e., the state transition matrix (STM) over precisely one period of the orbit, i.e., $\Phi(T, 0)$ [18]. Since τ approaches infinity for linearly stable orbits, when the monodromy matrix that possesses all eigenvalues equal to 1, the time constant is truncated to a maximum value of 5 when evaluating the stationkeeping benefit as,

$$\beta_{SK} = \min([\tau, 5]) \quad (7)$$

The goals for each mission are unique so each customer orbit benefit may be altered at the discretion of the designer; for this analysis, the mission specific benefit includes Lunar south pole coverage. The metric for measuring the south pole coverage is the proportion of the orbit period for which the elevation angle from the Lunar south pole is greater than 15° [21], i.e.,

$$\beta_{SP} = \frac{1}{T} \int_0^T \delta dt \in [\delta > 15^\circ] \quad (8)$$

Finally, the specific weighting values γ , are selected by the designer. For this assessment, more emphasis is placed on the customer orbits with south pole coverage, so the weighting values selected are $[\gamma, \gamma_{JC}, \gamma_{SP}, \gamma_{SK}] = [1, 1, 5, 1]$.

The orbit-selection criteria follow that of prior two-phase optimization and campaign-level trades in 2-body settings, extended to the cislunar regime where orbits and transfers are generated on demand for the current energy/phasing parameters rather than selected from a database [9, 10, 22]. This distinction is critical for OSAM in cislunar space. To assess the effect of subjective weights in Eq. (4), the values of $\gamma_{JC}, \gamma_{SP}, \gamma_{SK}$ are swept over the regions of interest, i.e. the L_1 and L_2 Halo and Lyapunov orbit families and the butterfly family. The sweeps displayed in Fig. 2 demonstrate broad regions where rankings are relatively invariant, except for the station keeping benefit which increases significantly for the NRHO members, indicating

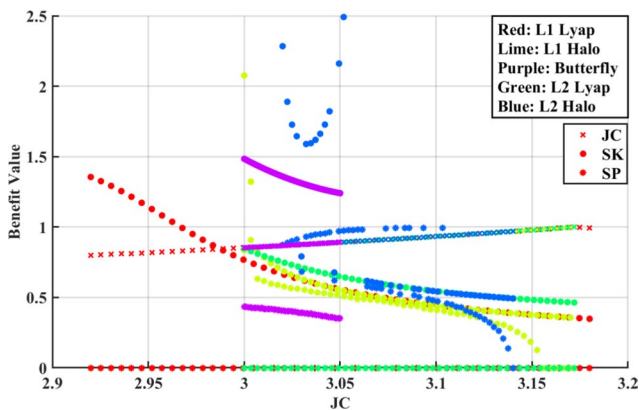


Fig. 2 Benefit-sensitivity panel showing benefit values across the L_1 and L_2 Halo and Lyapunov orbit families and the butterfly family. Families represented by color and benefit category represented by symbol

robustness to moderate reweighting. To prevent any single component from dominating, nominal weights are chosen, consistent with prior work, to keep representative orbits near unity in the aggregated score.

With this fitness score, a diverse set of networks are examined to determine the impact of the previously proposed criteria. Two types of simulations are noted here to identify priorities on this list and represent the broader analysis. In one set of simulations, the customer satellites are restricted to L_1 Lyapunov orbits and, alternatively, another set of simulations restricts customers to L_1 halo orbits. In the first example, ten different orbit networks are selected to identify optimal pairings between servicer and customer orbits. To limit the search space, customer orbits are restricted to L_1 Lyapunov orbits with Jacobi Constant (JC) values in the range of $JC = [2.92, 3.18]$, while servicer orbits are limited to L_1 orbits from the halo, Lyapunov, vertical, and axial families. For each simulation, two customer orbits are randomly selected within the defined JC range, and five servicer orbits are identified from random orbital families and JC values. All servicer orbits in the network are considered optional, that is, not all servicers are required for the simulation, while all customer orbits are mandatory. This formulation aids in identifying the servicers that best minimize the overall ΔV . The results appear in Fig. 3, that consists of six subplots. In the plots, 'x' marks the servicer orbits, solid colored dots represent customer orbits, and black dots indicate optional servicer orbits that were not included in the network. The color scheme is consistent across all subplots, that is, the network represented by a specific color in the "JC vs Time Constant" plot is the same as in the other plots. The six following subplots represent: a) each orbit JC vs Time Constant, b) JC vs ΔV for individual transfers corresponding to each orbit, c) ΔJC vs ΔV for each transfer, d) Period vs JC for each orbit, e) Period vs ΔV arriving and departing each orbit, and f) ΔJC vs Δ Period for each transfer.

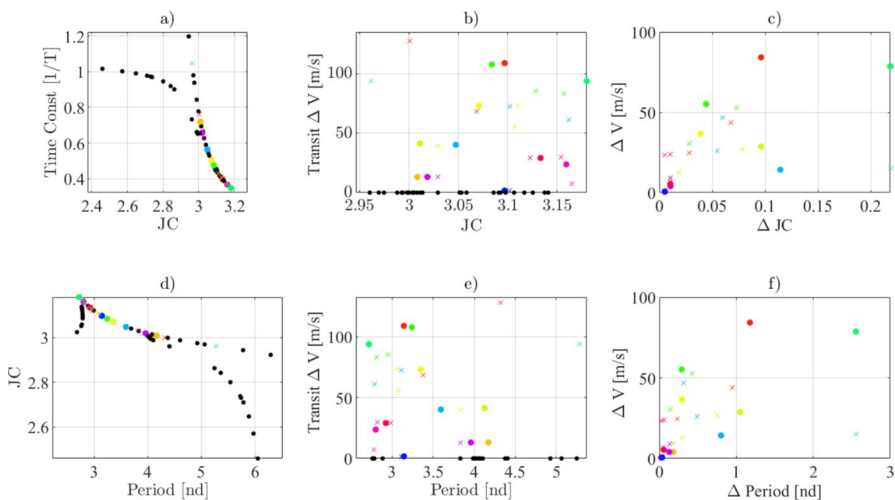


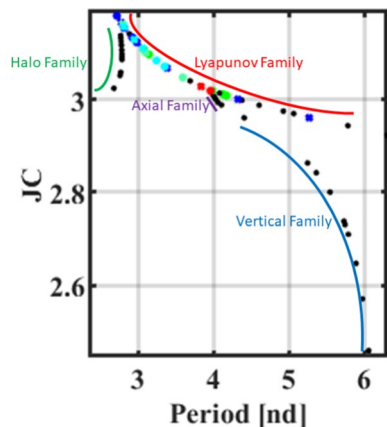
Fig. 3 Lyapunov customer network metrics. Each color represents individual networks 1-10, x's represent servicer orbits, and dots correspond to customer orbits

The preliminary simulations demonstrate that Lyapunov orbits outperform alternative periodic families as servicing hubs, a dominance reflected in the Period- JC map and the resulting ΔV comparisons. Upon examining the subplot in Fig. 3 of JC as a function of Period, the individual periodic orbit families are identified and labeled in Fig. 4. Note that, in every simulation that converged to an optimal schedule, only Lyapunov orbits served as hubs for servicer spacecraft (represented by the colored markers). In every scenario that attempts to implement servicer spacecraft in orbit families other than the Lyapunov family, the total ΔV that is required is significantly worse than using an alternative Lyapunov servicer. While this result may be expected because the Lyapunov family is planar and the other families are not, some of the halo options that are not selected are nearly planar with more comparable energy levels than the orbits that are actually exploited.

An inspection of the performance metrics in Fig. 3 reflect variations in orbital energy and geometry and the resulting impact on the ΔV required for servicing transfers. Examining subplots b) and e) in Fig. 3, there is apparently no strong correlation between the Jacobi Constant (JC) or period and the ΔV requirements. Contrary to prior expectations, the periods of the orbits does not significantly influence the ΔV necessary to achieve the proper phasing. This result is likely due to the inclusion of phasing revolutions in the solution space, where the servicer satellites remain in the customer orbit for some number of revolutions after completing servicing activities. This approach reduces the use of orbits with shorter periods that would otherwise increase the frequency of favorable configurations. In contrast, subplots c) and f) demonstrate a positive correlation between the changes in JC and period and the required ΔV for transfers. While there is a relationship between JC and the orbital period across orbit families, ΔJC demonstrates a stronger positive correlation with the required ΔV , indicating that variations in JC , i.e., energy, are more influential when determining transfer costs.

In the second simulation, all the initial orbit restrictions are the same with the exception that the customer orbits are restricted to L_1 halo orbits rather than Lyapunov trajectories. The resulting six comparison subplots for the second simulation appear in Fig. 5. Examining subplot d) for the plot of JC as a function of Period, it is evident that halo orbits are not the only servicers selected for the networks, in

Fig. 4 Lyapunov customer network Period vs JC with specific orbit families labelled



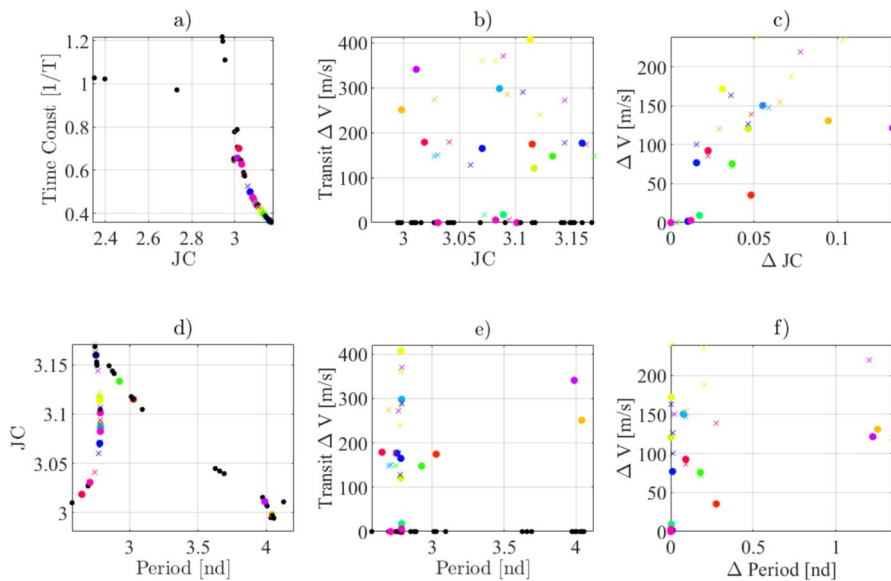


Fig. 5 Halo orbit customer network metrics. Each color represents individual networks 1-10, x's represent servicer orbits, and dots correspond to customer orbits

contrast to the Lyapunov network where only servicers within the same family are selected. However, as apparent in the other subplots representing the required ΔV , networks incorporating servicers from different families required the largest ΔV . The only networks that included out-of-family servicers and required less than 150 m/s per transfer involve a servicing hub in a Lyapunov orbit with a nearly-planar halo customer orbit. Consistent with the Lyapunov orbit simulation, the JC and period of the orbits offer minimal impact on the ΔV requirements, while ΔJC demonstrates the strongest correlation with the required ΔV .

For both simulations, the impact of the time constant on both transfer ΔV and the required transfer time is also examined. Comparisons between the total ΔV required and the transfer time versus orbit time constant for each individual transfer in both networks are plotted in Fig. 6. In the figure, the x-axis represents the time constant for the originating orbit in each transfer, while the y-axis illustrates the required ΔV and the transfer time. The plot reveals that the time constant actually incurs minimal impact on these transfer parameters. Note that all orbits in these simulations are linearly unstable, with time constant values less than 1.2 revolutions. If stable or nearly stable orbits, such as Near Rectilinear Halo Orbits (NRHOs), are incorporated, the time constant is expected to introduce a more significant effect on transfer design.

The two simulations highlight the relative significance of the previously identified criteria in selecting orbits for on-orbit servicing networks in the cislunar domain. Firstly, while symmetry is not directly measurable, the most significant factor in orbit selection is the geometric similarity of the orbits within the same family. This requirement, though restrictive, results in the lowest ΔV costs for transfers between orbits in the same family. Secondly, as expected, the difference in the Jacobi Constant (JC) value between two orbits offers the strongest correlation with the ΔV

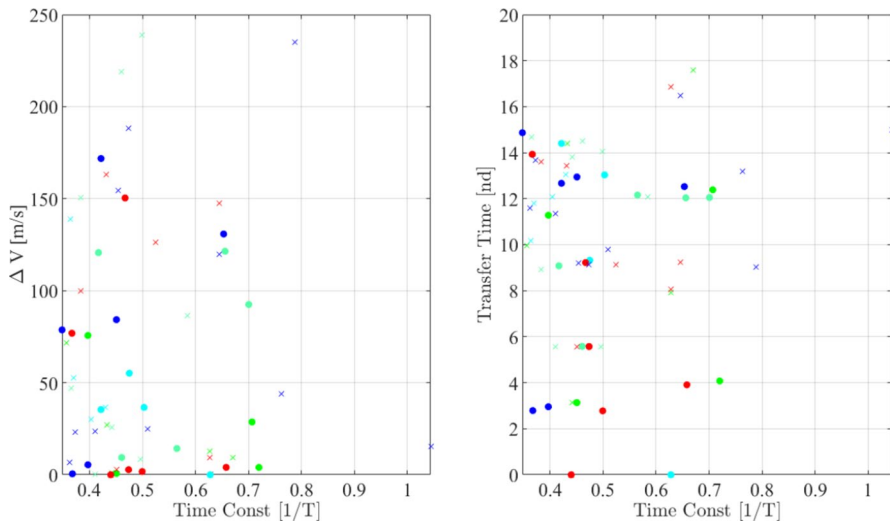


Fig. 6 ΔV and transfer time vs time constant for transfers throughout both networks. Each color represents individual networks, xs represent servicer orbits, and dots correspond to customer orbits

costs. Specifically, for transfers within a single family, the energy gap, i.e., ΔJC , serves to correlate directly with a rise in the propulsive requirements. The correlation between ΔV and the energy gap also demonstrates a relationship with a difference in the orbital periods, but further characterizing the effect of a variation in period is challenging due to the interdependence between JC and orbit period. Finally, the stability and individual periods for the orbits yield minimal impact on the required ΔV and transfer time. Since these simulations are implemented with linearly unstable orbits, a more noticeable relationship is expected if stable or nearly stable orbits are incorporated in future simulations. Therefore, when selecting orbits for satellite servicing networks in cislunar space, the previous criteria is limited to two dominating factors: similar JC values, indicating comparable energy levels; and selecting orbits with symmetric manifold geometries, most realizable in the form of orbits belonging to the same family.

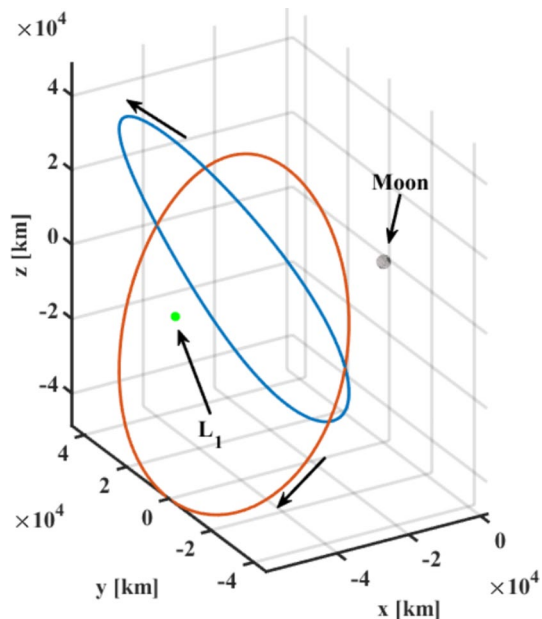
4 Transfer Generation

Since manifold trajectory arcs represent impulse-free departures from and arrivals into an operating orbit, they are an attractive option for construction of transfers between orbits. In an ideal transfer scenario, there exists an unstable manifold trajectory arc that departs from the original orbit and perfectly links with a stable manifold trajectory arc flowing into a desired target orbit. Such a scenario assumes both orbits are unstable in a linear sense. In this scenario, assuming that the original and target orbits exist at the same value of JC , a free transfer, termed a heteroclinic connection, exists between the two orbits. One technique for identifying a potential heteroclinic connection originates with the identification of two orbits at the same energy level.

The following example demonstrates a transfer between a northern Earth-Moon L_1 halo orbit and a southern L_1 halo orbit near $JC = 3.07$, both orbits are linearly unstable and plotted in Fig. 7. To construct a heteroclinic connection between two periodic orbits at the same energy (JC) level, the following steps are successful. The unstable manifold trajectory arcs from the northern halo orbit and the stable manifold trajectory arcs from the southern orbit are first propagated, forwards and backwards, respectively, to a predefined hyperplane to create a Poincaré map. Every intersection of a trajectory with the hyperplane (Σ) is represented as a single point on the map. The points are then viewed on the 2D map to identify potential connections. In addition, glyphs offer the opportunity to represent additional velocity information at the hyperplane crossing, providing more insight into potential linkages and behaviors at the intersection points [23]. Selection of the hyperplane is largely the preferences of the designer, however, useful hyperplanes include: the $y - z$ plane at the Moon's x position value, as well as the $x - z$ plane, and the $x - y$ plane. The northern to southern transfer exploits $x - y$ symmetry, so the $x - y$ plane is selected as the hyperplane in this example.

Since the halo orbits and their manifolds possess multiple crossings on the selected hyperplane, multiple Poincaré maps identify an expanded collection of potential connections. By viewing each crossing on the hyperplane as an individual map, better initial guesses leading to heteroclinic transfers are potentially available. In Fig. 8, the red and blue points represent the crossings of the unstable and stable manifold arcs, respectively, and the solid circles indicate the closest connections in the full six-dimensional state, including velocity. It is clear that the closest points to a heteroclinic connection between the two halo orbits occur at the fourth crossing of this hyperplane because the red and blue curves of points have an intersection with similar velocity components.

Fig. 7 Example L_1 Halo Orbits Moon-Centered CR3BP: blue orbit is the northern counterpart and red orbit is the southern counterpart



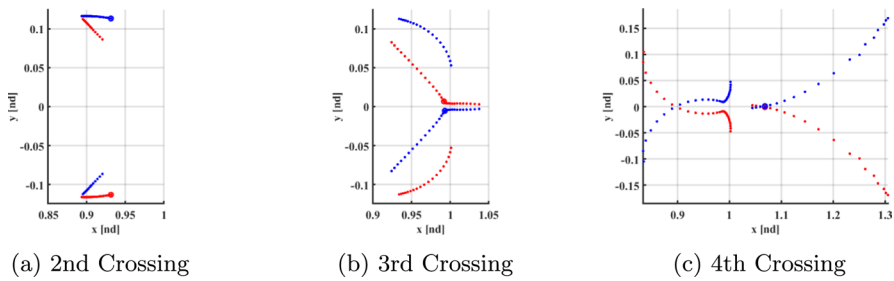


Fig. 8 L_1 northern and southern one-sided manifold crossings of $\Sigma(z = 0)$. Blue: crossings of stable manifold trajectories from southern halo orbit; Red: crossings of unstable manifold trajectories from northern halo orbit

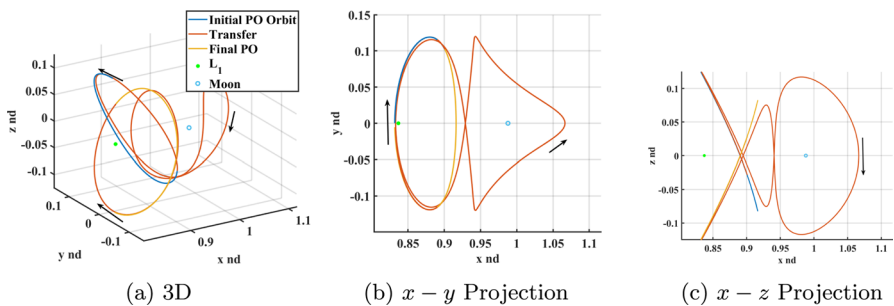


Fig. 9 Views of Heteroclinic Connection Between Halo Orbits CR3BP. Blue: Northern halo orbit; Red: N-S transfer; Yellow: Southern halo orbit

In state space, this initial guess corresponds to a position error of 750 km and a velocity error of $7.5 \frac{m}{s}$, relying on a manifold step-off distance of 80 km. After iteratively reducing the ΔV using slack variable constraints, the transfer is converged for full state continuity between the manifold trajectory arcs with no time discontinuity. During the convergence processes, the initial states along the periodic orbits are free variables, allowing both orbits to shift throughout the corrections scheme. With this variance, the initial and final orbits are both adjusted from the original values $JC = 3.07$ to $JC = 3.0723$ and yield an impulse-free transfer from the northern halo orbit to the southern halo orbit. Due to symmetry in the CR3BP, a reflection of this transfer supplies the impulse-free transfer from the southern halo orbit to the northern halo orbit. The fully converged heteroclinic connection appears in Fig. 9.

While heteroclinic connections are the ideal links for generating transfers between orbits at the same energy level, they are not always available between every orbit pair, even a pair at the same energy level. However, the process for using manifold trajectory arcs to generate an initial guess, and to iteratively reduce the ΔV using slack variables, is useful for identifying low-cost transfers between orbits of interest as well. Plotted in Fig. 10 are the manifold trajectory arcs from a butterfly orbit, an L_1 halo, and an L_2 halo orbit that are employed for initial guess transfer generation [24]. The manifold trajectory arcs in Fig. 10 are not converged to serve as a free transfer, therefore, they include both velocity and position discontinuities at the $z = 0$ hyper-

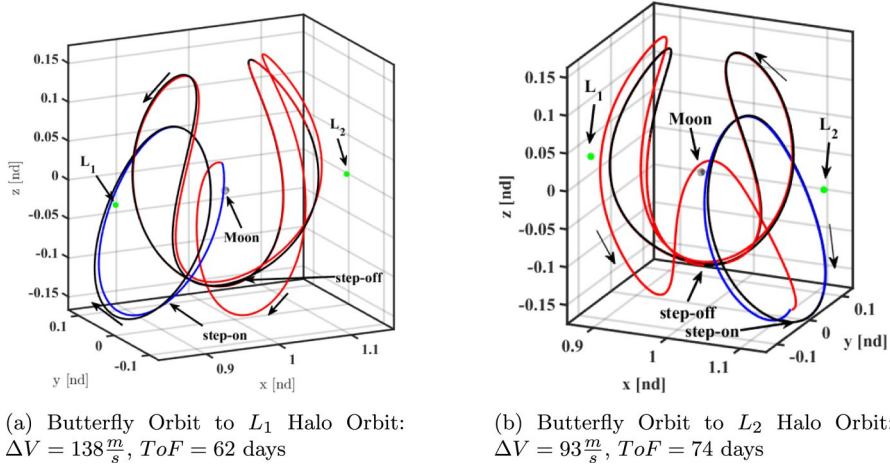
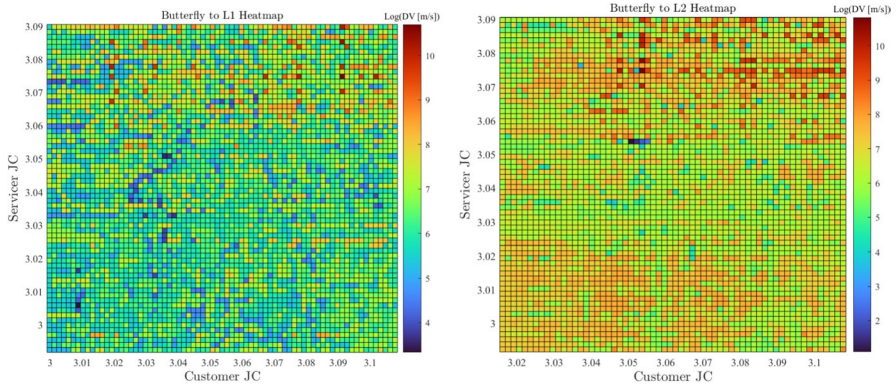


Fig. 10 Manifold trajectory arc initial guesses: Black - underlying periodic orbits, Red - Unstable manifold trajectory arc, Blue - stable manifold trajectory arc

plane. These manifold trajectories emerge as a relatively fast baseline estimate for the transfer costs and transfer times required to reach each of the customer orbits. In Fig. 10, the ΔV corresponds to the velocity discontinuity between the manifold arcs at the hyperplane crossing and the time of flight (ToF) is evaluated in terms of the following quantities, i.e., $ToF = \Sigma(\tau_{butterfly}, \alpha_1, \alpha_2, \tau_{halo})$. Where flight time is the summation of the time along the following 4 phases: the butterfly orbit from the left-most x -axis crossing to the step-off location, the unstable manifold trajectory arc from the butterfly step-off location to the hyperplane crossing associated with this initial estimate, the stable manifold trajectory arc from the hyperplane crossing to the halo orbit step-on location, and the final halo orbit from the step-on location to the x -axis crossing associated with apolune.

Similar to the process for locating heteroclinic connections, the energy levels for the initially selected servicer and customer orbits are varied to determine combinations of orbits that offer generally lower ΔV requirements. The energy levels for each orbit are varied in steps of $JC \pm .0025$, then, manifold trajectory arcs are propagated to a desired hyperplane, and the estimated ΔV for the closest connections are retained. For the L_1 halo orbits, the .0025 variation in JC value corresponds to a change in the maximum z -excursion of about 900 km. The estimated ΔV for all energy level combinations is then plotted in the form of a heat map to visualize nearby orbits that offer solutions with potentially lower propellant requirements. The heat maps plotted in Fig. 11 represent the estimated ΔV values to transfer between nearby butterfly orbits and halo orbits. The heat maps are constructed using the manifold trajectory arcs for the respective orbits, i.e., those similar to the orbits in Fig. 10. In Fig. 11, the range in ΔV that is required to transfer from the butterfly orbit to orbits in either of the halo orbit families is largely similar. Examining the minimum ΔV initial guesses from the maps provides the transfer options plotted in Fig. 12.

Each heatmap offers an estimate of the ΔV cost required to transfer between two orbits, highlighting the initial geometries that may enable the most efficient ΔV trans-



(a) Transfer cost heat map for butterfly orbits in range $JC = [2.99, 3.09]$ to L_1 halo orbits in range $JC = [3, 3.11]$. (b) Transfer cost heat map for butterfly orbits in range $JC = [2.99, 3.09]$ to L_2 halo orbits in range $JC = [3.01, 3.11]$.

Fig. 11 Estimated Transfer Cost Heat Maps for Butterfly Orbits to L_1 and L_2 Halo Orbits

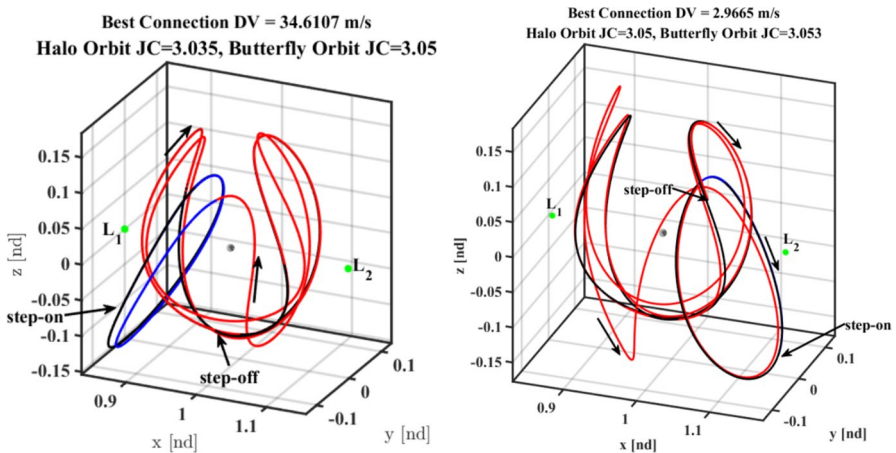


Fig. 12 Lowest ΔV options from a butterfly orbit to L_1 (left) and L_2 (right) halo orbits (CR3BP)

fers. With these preliminary estimates, the actual transfer trajectories are refined and converged in the CR3BP using a differential corrections scheme, ensuring accuracy and feasibility. Each cell of the heatmap represents various transfer options between two desired orbits and forms the backbone of the transfer generation process.

4.1 Phasing Constraints

The most critical aspect in designing a servicing network is ensuring that the servicer satellite reaches the customer orbit with the proper timing for rendezvous. As an example of the influence that transfer time places on a solution, recall the transfer from the butterfly orbit to the L_1 halo in Fig. 12. Converged for position continu-

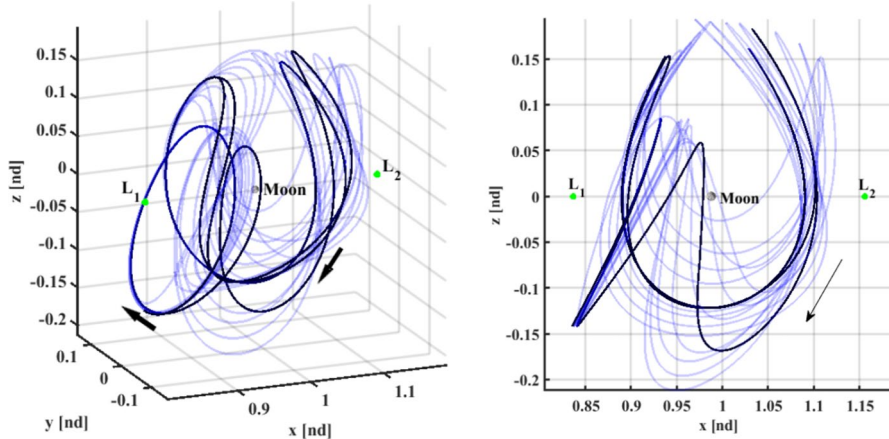
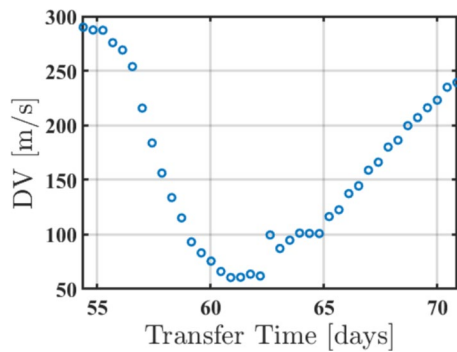


Fig. 13 Butterfly to L_1 transfer family CR3BP, Black trajectory: baseline solution

Fig. 14 ΔV as a function of transfer time for butterfly to L_1 transfer family



ity, the solution requires a ΔV roughly equal to $100 \frac{m}{s}$ while maintaining the transfer time at 62 days. With the converged solution, natural parameter continuation is employed to vary the transfer time and to supply a family of transfers that originate from the same butterfly orbit and reach the same L_1 halo orbit. The family of transfers is plotted in Fig. 13 where the black transfer is the original baseline from the heat map solutions. The ΔV and time of flight (ToF) for each of the transfers in the family appears in Fig. 14. A locally minimum ΔV solution for this initial guess is observed and any change in the transfer time increases the required ΔV .

The relationship between the required ΔV and the ToF that yields a minimum cost continues across various transfer scenarios and is evident as this process is repeated across other transfer families. The transfers across 10 different families from the butterfly orbit to the L_1 halo orbit appear in Fig. 15 where the results are represented in terms of ΔV cost as a function of ToF. Each family is based on a different initial solution from Fig. 11 and continued across a range of transfer times. The results from Fig. 15 support the fact that the selected initial guess from the heat maps clearly impacts the solution space and the family evolution. To highlight this impact, note that in Fig. 15, six different initial guesses yield transfer families where at least one

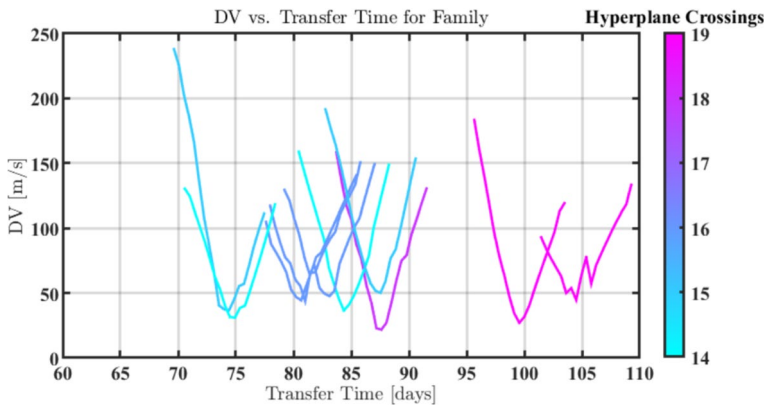


Fig. 15 Butterfly to L_1 Transfer Families Generated from the Heat Map; Colored by Hyperplane Crossings

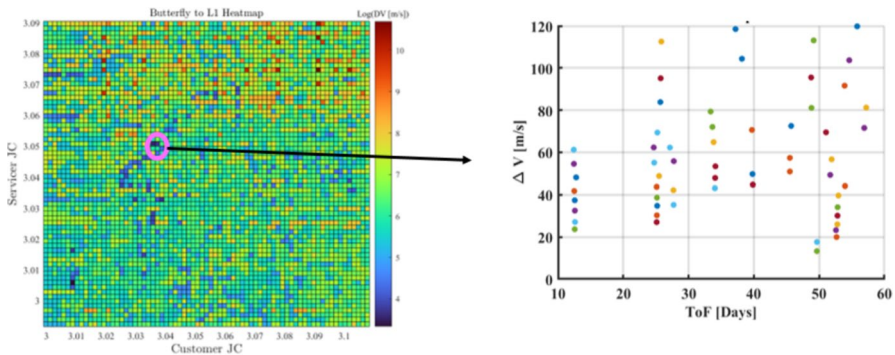


Fig. 16 Sample heatmap for transfers from a butterfly orbit to an L_1 halo orbit (Left) and ΔV as a function of ToF transfer options for selected energy configuration (Right)

member renders a transfer time of 85 days (6 separate lines cross the $X = 85$ days axis), yet one clear solution requires the lowest ΔV among all the options (the transfer with the fewest hyperplane crossings). Therefore, it is clear that identifying suitable initial guesses that satisfy phasing constraints is critical (Fig. 14).

To select the lowest ΔV initial guess for a desired transfer time, and produce various network combinations, a linear interpolation scheme that builds off of the heatmap is introduced. As visualized in Fig. 16, each cell in the heatmap represents various transfer options between two desired orbits.

The estimated cost and transfer time is retrieved from the heat map data for the most appropriate solutions that satisfy initial ΔV and transfer time constraints (Fig. 16). Since the heat maps are constructed from manifold arcs for each orbit, it is only approximate, but still delivers insight for a successful initial guess. For each transfer in the network, the ΔV and the transfer time are combined to construct the cost for spacecraft i to transfer to target j .

$$b_{ij} = \Delta V_{ij} + .15 \Delta t_{error} \quad (9)$$

As opposed to other strategies that accommodate only the ΔV or the transfer time, this scheme incorporates the phasing error between the initial transfer and the target spacecraft. The non-dimensional timing error is multiplied by a weighting value equal to $.15 \frac{km}{s * t_{nd}}$ to weight the cost of altering the original transfer time in terms of ΔV , therefore yielding a total cost b_{ij} in $\frac{km}{s}$. The weighting value of .15 is selected because it aligns well with the average slopes observed in Fig. 15, but it is easily altered to match the needs of the designer. The coefficient in Eq. (9) is selected post-testing on the transfer families used in this study and is found to be stable for those families, yielding consistent schedules without retuning. This value is problem-dependent, reflecting how ΔV trades against time-of-flight for a given transfer class and geometry; different families (e.g., halo-to-halo variants, planar vs. vertical manifolds, direct vs. stitched arcs, differing energy and phasing) can exhibit different local $\frac{\delta(\Delta V)}{\delta ToF}$ behavior. Accordingly, the present coefficient should be read as a practical surrogate for the networks analyzed, rather than a universal constant. The authors are actively investigating weighting schemes that cluster transfer families and assign family-/geometry-conditioned phasing weights, as well as parametric or learned surrogates that map family and geometry to κ ; this generalization is left to future work.

The following phasing-informed transfer design strategy is summarized as follows: given the initial and final orbits, manifold arc trajectories are incorporated to construct a heatmap for potential transfers between the orbits. Then, the ΔV and transfer time from the heatmaps, in combination with the current phasing, aid in the selection of the heatmap initial guess that minimizes the estimated cost. Finally, the transfer initial guess is converged in a differential corrections scheme that leverages slack variables to reduce the ΔV discontinuity and ensure the proper phasing for rendezvous. This approach yields a phasing-informed strategy for designing trajectories throughout servicing networks without the need to converge hundreds of transfer options. This approach also supplies an improved estimate of the true cost for a converged transfer when the underlying ToF must be varied to fit phasing constraints.

5 Servicer Scheduling

As with orbit selection and transfer generation, the task of designing schedules for servicer satellites in OSAM networks presents a vast search space with large variations in potential solutions. Even when the ordering of customers visited by servicers is fixed, the number of solutions for a small network comprised of only a few orbits can quickly scale to millions of options without even allowing reordering of the schedules. Therefore, two solutions are proposed to address the challenge of efficient scheduling while also maintaining dynamical insights.

5.1 Auction Algorithm

It is cumbersome to manually track the phasing for each spacecraft and to identify minimum ΔV solutions when designing a servicing network that involves many customers and servicers. Therefore, an auction-based scheduler (AA) as a fast allocation method under dynamics-consistent, phasing-aware edge costs is evaluated. The standard AA addresses assignment under on-demand transfer costs and is used strictly as an allocator, not as a tour-construction algorithm. It scales efficiently to large instances and provides schedule allocations when rapid turnaround is paramount. While the AA offers rapid allocations with low computational burden, the network problem considered here is an OTSP with time- and order-dependent costs rather than a pure assignment problem. While the standard AA is intended for pure assignment problems, a recursive AA is implemented due to its computational efficiency, similar to previous debris removal work [7]. When a full tour is required and ground computation is available, a separate GA-based tour optimizer is also proposed.

The AA is demonstrated using a network comprised of one servicing satellite originally in a butterfly orbit, one servicer originally in a P2HO2, and three customers in halo orbits, however, the auction approach is scalable to any number of servicer/customer satellites in any orbits. The customer orbits include L_1 and L_2 halo orbits, each with $JC = 3.05$ and an L_2 NRHO characterized by $JC = 3.05$ and a period of 6.4 days. The two servicer craft are originally placed in a northern butterfly orbit (P2HO1) [25] corresponding to a $JC = 3.05$ and period of 22.4 days and a P2HO2 at $JC = 3.05$. Two updates to the standard auction algorithm [26] are implemented. The first is an additional value added to the cost function to model the cost associated with improper phasing. This addition is essentially an estimate for the ΔV necessary to alter an underlying transfer such that the transfer time matches a desired value that is required for rendezvous. The second modification is the addition of a penalty for the cost to return to the servicing hub in terms of ΔV . If return costs from a customer orbit to the servicing hub orbit are high, this penalty prevents the servicer associated with that hub from being assigned to that customer.

5.1.1 Cost and Expense

The success of an auction is dependent on accurately assessing the costs linked to each spacecraft-target pairing, ensuring that the ultimate assignment aligns with a formation that is favorable from the operator perspective. The cost for each servicer to reach a customer orbit employs the estimated ΔV and the transfer time from the heat map solutions, augmented with the cost to return to the servicing hub and some small expense representing the latest serviceability cost. First, the estimated cost and transfer time is retrieved from the heat map data for the most appropriate solutions that satisfy initial ΔV and transfer time constraints. Since the heat maps are constructed from the manifold arcs for each orbit, it is only approximate, but offers insight for a successful auction algorithm. For each spacecraft in the network, the ΔV and the transfer time are combined to construct the cost for spacecraft i to transfer to target j consistent with Eq. (9). As the auction algorithm continues, the price for each customer spacecraft to be serviced is modified. Therefore, the total expense for

spacecraft i to reach target j (ν_{ij}) is a combination of the cost (b_{ij}), price (p_{ij}), and an extra penalty (b_{return}) that represents the cost for the servicer to return to its original servicing hub orbit, i.e.,

$$\nu_{ij} = b_{ij} + p_{ij} + b_{return} \quad (10)$$

The additional return cost is added to avoid the scenario of a servicer cornering itself through multiple low ΔV inter-network transfers and being left with a prohibitively large ΔV cost to return to the servicing hub.

5.1.2 Bidding and Assignment

Once the costs are estimated for all servicer satellites to reach all other orbits in the network, the auction begins. Initially, all spacecraft (servicers) are unassigned. As the auction progresses, spacecraft are incrementally assigned until the auction concludes, with each spacecraft being allocated to a target orbit. Each auction round commences with the bidding phase, wherein all servicer spacecraft are eligible participants, in contrast to some algorithms that only allow unassigned spacecraft to participate. Each available servicer places a bid for the target with the lowest expense. The bid γ_i for target j_i is created by using the difference between the lowest expense ν_{ij} and the second lowest expense w_{ij} . Note that ν_{ij} is the total expense from Eq. (10). Alternatively, the inverse of the minimum expense is more advantageous at times and is the strategy that is implemented here,

$$\gamma_i = w_{ij} - \nu_{ij} \quad \text{OR} \quad \gamma_i = \frac{1}{\nu_{ij}} \quad (11)$$

With one bid placed per servicer, every target that received a bid follows up with a procedure to be assigned a spacecraft. First, the maximum bid γ^* and the associated bidder i_j is noted. Next, the price for the customer, as originally input into Eq. (10), is updated accordingly, i.e.,

$$p_j^+ = p_j^- + \gamma^* \quad (12)$$

This price increase (γ^*) prevents the same target from being serviced twice without removing them from the bidding process. After all targets that have received bids undergo the assignment of a servicer, the algorithm reverts to the bidding phase. After a servicer is assigned to a customer, it is assumed, after completing the required transfer, that the services are performed over one revolution of the customer orbit. The cycle repeats if there are any targets that remain unassigned. In addition to tracking pricing updates, the auction algorithm also must track the current phase for all of the satellite instantaneous locations. The tracking phase is an important addition because the baseline transfer between two orbits in the network changes over the course of the simulation due to imperfect periodic resonances. The auction concludes when every customer spacecraft is assigned, and the servicer satellites have returned to their respective servicing hubs, although not necessarily all at the same time. This

auction framework is fairly straightforward and accommodates scenarios involving an uneven distribution of spacecraft and targets, a likely scenario when servicing near the Moon. A sample auction scenario is summarized in Fig. 17. In round 1 in the table, each servicer satellite (one located in a butterfly orbit and the other in a P2HO2, both characterized by $JC = 3.05$) places a bid for one of the customer satellite orbits (located in an L_1 halo orbit, an L_2 halo orbit, and an L_2 NRHO) corresponding to the target orbit with the lowest associated ΔV transfer cost. In round 1, both servicers placed bids for the L_1 halo orbit customer because this customer had the lowest cost for both servicers. At the end of round 1, the servicer 1 satellite is assigned to the L_1 halo orbit and the transfer time is updated for phasing management. In round 2, both servicing spacecraft place bids to separate customer orbits, so each gets assigned accordingly. In round 2, Servicer 1 does not have an associated cost to service the L_1 customer because it is already in that customer orbit. Since both servicers in round 2 have different customers that offer the lowest individual costs, they place bids on different customers. At the end of round 2, no customer satellites remain, so the auction concludes. The final schedule includes the servicer 1 satellite providing services to the L_1 halo orbit customer, then to the L_2 NRHO customer and the servicer 2 satellite only providing services to the L_2 halo orbit customer. Both servicers return to the hub after the assigned tasks.

Since the auction algorithm incorporates a significant phasing penalty, a servicing schedule is generated with significantly lower ΔV costs than any pre-determined hand-picked transfer networks. In another example, a single servicer satellite in a butterfly orbit offers services to three customers in the previously noted halos, the costs and transfer times that are required for all servicing sequences are compared in Fig. 18. Each line in the figure represents the evolution of the ΔV and time required to service the network. A step change in the slope of the line represents a ΔV impulse applied to transfer to the next customer in the network. The final coordinates for the lines represent the total cost to service each customer before returning to the initial servicing hub. In this example, the servicer is allowed to return to the servicing hub mid-schedule to resupply or for use as a phasing orbit. It is assumed that the servicer

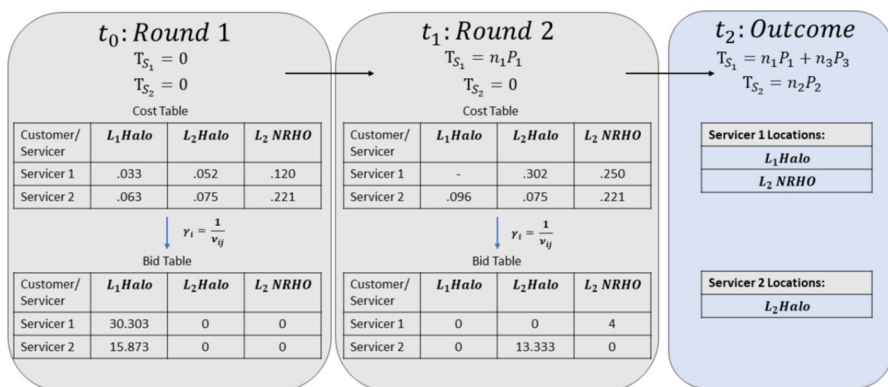


Fig. 17 Example of auction algorithm assignments. Servicer 1: butterfly orbit servicer, Servicer 2: P2HO2 servicer, Customer 1: L_1 halo orbit, Customer 2: L_2 halo orbit, Customer 3: L_2 NRHO

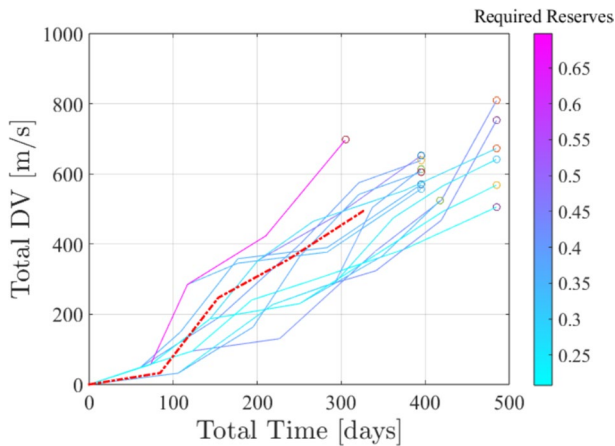


Fig. 18 Total ΔV as a function of ToF for various servicing sequences with the one selected by the auction algorithm in red and dashed

refuels while momentarily at the hub, so the colorbar represents the maximum ΔV [$\frac{km}{s}$] reserves required by the servicer spacecraft to perform all services. Therefore, when the servicer does not stop at the hub mid-schedule, the ΔV amount required in reserves is typically larger. Noticeably, the schedule and the transfers selected by the auction algorithm outperform other schedules, based only on minimum ΔV candidate solutions from the heatmaps, in both total ΔV required and the total time required to accomplish all transfers. The schedule selected through the auction process follows butterfly hub to L_1 halo customer, then the L_2 NRHO customer, then the L_2 halo customer before returning to the hub in the butterfly orbit.

While the auction algorithm offers rapid solutions with a reasonably low computational requirement, it lacks insight into the transfer architecture further into the network. While there are multiple variations of auction algorithms, the original basic framework solves assignment problems and network optimization problems. The AA is an iterative method that locates optimal prices and assignments that maximize the net benefit. However, the OSAM application is a version of the travelling salesman problem rather than pure assignment. With the travelling salesman problem, the goal is the minimum cost path for a single salesman to visit a set number of nodes before returning to the starting location. With an interwoven network of nodes, the goal is to find the path that hits all of the nodes with the least travel costs. Therefore, when ground computation is available, a genetic algorithm offers more robustness than the AA.

5.2 Genetic Algorithm

A Genetic Algorithm (GA) is an optimization technique inspired by the principles of natural selection and genetics, often used to solve complex problems with large search spaces [1]. The algorithm initiates by randomly generating an initial population of candidate solutions, labelled individuals or chromosomes. Each chromosome represents a possible solution to the problem, typically encoded as a binary string. A

fitness function is then defined to evaluate each chromosome for solving the problem, with the fitness score indicating the quality of the solution.

Once the population is initialized, the algorithm enters a loop of selection, crossover, and mutation to evolve the population over successive generations. During the selection phase, individuals are identified based on their fitness scores, those with higher fitness more likely to be selected. The selection process ensures that the best solutions are more likely to pass their genetic material to the next generation. Common selection methods include roulette wheel selection, tournament selection, and rank-based selection. In the crossover phase, selected individuals (parents) are paired to produce new offspring through a recombination process. During crossover, segments of the parents' chromosomes are exchanged to create new combinations of genes, that represent potential solutions. Various crossover techniques, such as single-point crossover, multi-point crossover, and uniform crossover, are options depending on the problem requirements. Mutation then introduces small random changes to individual chromosomes, ensuring genetic diversity within the population and allowing the algorithm to explore new areas of the solution space. The mutation rate is usually kept low to maintain a balance between exploration and exploitation. Newly created offspring, along with some of the best individuals from the previous generation, form the next generation of the population. This process repeats for a set number of generations or until a satisfactory solution emerges. Over time, the population is expected to converge towards an optimal or near-optimal solution as the genetic algorithm explores the search space. The key components of the GA include population initialization, fitness evaluation, selection, crossover, and mutation that work together to iteratively improve the quality of solutions, rendering a tool for solving complex optimization problems.

The GA is proposed for the advantages of adapting to complex, scalable search spaces, and most importantly, flexibility in representation. The genetic algorithm is typically used to solve optimization problems that are not well-suited for standard optimization algorithms, including problems where the objective function is discontinuous or nondifferentiable. As opposed to other approaches, the GA offers the ability to represent important considerations such as optional nodes in a trade space analysis.

5.2.1 Implementation

As opposed to the original Traveling Salesman Problem (TSP), the Orbit Traveling Salesman Problem (OTSP) [5, 6], when implemented near the Moon, renders a distinctive set of challenges. Considering N nodes on a graph, each pair of nodes is defined with distance $d_{i,j}$ between them. In a standard TSP, the goal is to determine an ordering of the nodes π that minimizes the total traveled distance

$$\min J_{\pi} = \sum_{i=1}^{N-1} d_{\pi_i, \pi_{i+1}} \quad (13)$$

by a single salesman. When implementing the OTSP, many of the standard assumptions in the TSP break down. The most impactful difference between the TSP and OTSP is the time dependency inherent in transferring between orbits, especially in the case of OSAM where rendezvous is required. The time dependency directly impacts the ΔV requirements for a single transfer, where the ΔV required to transfer between two orbits at one epoch may or may not be the same at a different epoch, that is, $\Delta V_{i,j}(t_1) \stackrel{?}{=} \Delta V_{i,j}(t_2)$. This time dependency leads to a break in the path dependency assumption in the TSP. Specifically, the cost to transfer to a customer orbit, is not necessarily more expensive when an intermediate stop is included, i.e., $\Delta V_{1,2} + \Delta V_{2,3} \leq \Delta V_{1,3}$. In addition, there are also an infinite number of transfer paths that exist between two orbits given an initial epoch. To simplify the optimization process, a two-step solution is introduced: (i) employ a search algorithm to determine the minimum cost to transfer between two orbits; and (ii) use a second GA process to determine the servicing order. As visualized in Fig. 19, given an initial set of servicer and customer orbits, the GA initializes the first generation of n random chromosomes representing various servicing schedules.

Then, each member of the generation is individually evaluated for its fitness score. The chromosome is divided into pairs of genes representing orbits and the required ΔV is estimated using previously determined transfer options. The transfer option that minimizes the ΔV , consistent with Eq. (9), is determined for the given epoch and passed back to the evaluation process. Once the fitness score for all the chromosomes in the generation is determined, several crossover operations are applied to manipulate the next generation. This evaluation and crossover process is repeated for m generations or until a satisfactorily optimal solution is determined.

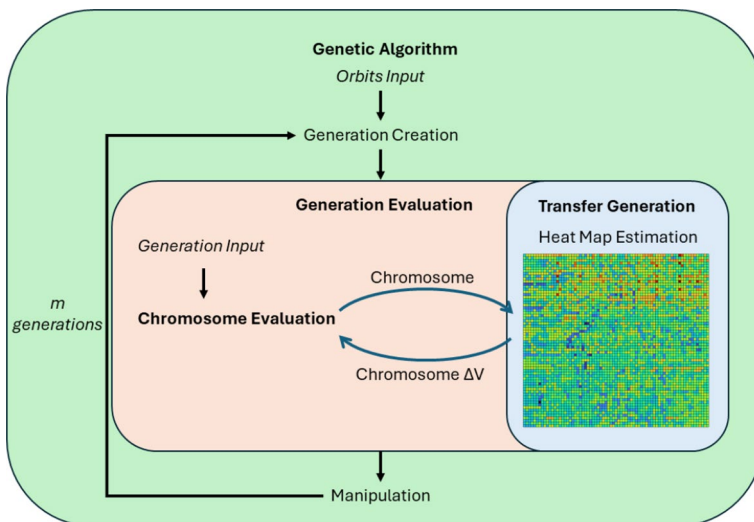


Fig. 19 Genetic algorithm systems with heatmap transfer evaluation. High level GA process corresponds to (ii) and heat map estimation corresponds to (i)

5.2.2 Chromosome Representation

To apply a Genetic Algorithm (GA) to any optimization problem, it is essential to represent solutions as valid chromosomes, ensuring that the crossover operation yields valid chromosomes as well. In contrast to the usual binary bit representation of the solution space, for the Multi-Objective Traveling Salesman Problem (MTSP), three main chromosome representation methods are commonly employed: the one-chromosome method [27], the two-chromosome method [28], and the two-part chromosome method [29, 30]. The representation here involves one-chromosome with artificial depots, to represent servicer starting locations, similar to that of Al-Furhud [31], but includes the addition of three components that allow for the incorporation of optional customer orbits and phasing revolutions. A sample chromosome for two servicer spacecraft and six customer satellites, $n_{serv} = 2$ and $n_{cust} = 6$, appears in Fig. 20. In the sample: each integer value less than n_{cust} , but greater than zero, represents a customer orbit (Blue); each value greater than n_{cust} , but less than $n_{cust} + n_{serv} + 1$, represents a servicer (Red); each zero (Black) value represents the servicer remaining in the previous orbit for one additional revolution to alter the phasing with future orbits; each value greater than $n_{cust} + n_{serv}$ represents an optional customer orbit that may or may not be included in the final solution (Purple); all values following the termination genome, -1 , are not included in the schedule (Gray). Should a chromosome represent *not* using/servicing the optional orbit, it is placed after this termination genome.

In this chromosome, [7, 2, 3, 0, 6, 8, 0, 9, 5, 4, 0, 1], there are 9 genes including an artificial depot, 8, and an optional orbit 9. This sequence represents the tour of two servicers, 7 and 8, following the path of $7 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 7$ and $8 \rightarrow 9 \rightarrow 5 \rightarrow 4 \rightarrow 1 \rightarrow 8$, respectively. Servicer 1 departs from orbit 7, visits customers 2 and 3, conducts one phasing revolution in orbit 3, visits customer 6 and then returns to the original servicing hub in orbit 7. Servicer 2 initiates a sequence in orbit 8, immediately remains for one phasing revolution, visits customers 9, 5, and 4, conducts a phasing revolution in orbit 4, visits customer 1, and then returns to its original servicing hub in orbit 8. This chromosome representation enables modeling of complex sequences involving optional orbits and phasing revolutions better than a typical binary representation. It also ensures that crossover strategies produce valid chromosomes.

5.2.3 Chromosome Manipulation

After evaluation across one generation in the Genetic Algorithm (GA), several manipulation techniques are applied to produce the next generation of solutions. The first step is selection, where individuals from the current population are identified based on their fitness scores. This process favors individuals with higher fitness, giv-



Fig. 20 Example Chromosome: Red - servicer start, Blue - customer orbit, Black - phasing rev, Purple - optional customer, Grey - terminated

ing them a better chance of contributing their genetic material to the offspring. Rank-based selection is employed to ensure that better-performing solutions are more likely to be selected. Following selection, the crossover, or recombination, operation is accomplished. Crossover combines the genetic information from two parent individuals to create new offspring, allowing the algorithm to explore new areas of the solution space. Mutation is another key manipulation technique, where small random changes are introduced to individual chromosomes. By altering one or more genes, mutation aids in maintaining genetic diversity within the population and prevents the algorithm from converging prematurely to suboptimal solutions. Finally, elitism is incorporated into the GA to preserve the best-performing individuals from the current generation. By carrying over these elite solutions directly to the next generation without any modifications, the algorithm ensures that the best traits are retained and not lost during the evolutionary process.

Selection: For the selection and ranking of the chromosomes, the fitness score for each individual is evaluated. Since the needs of each design effort are different, priorities on the weighting of fitness score components may also vary. For this investigation, the ΔV estimate from the transfer generation process plays a large role in the fitness score evaluation as displayed in Eq. (4). When the underlying orbits are fixed (not allowed to shift in the corrections procedure), the trailing two parts of the fitness score are omitted (servicer count penalty and customer benefit) so that the resulting schedule is only dependent on the total ΔV_T required.

Greedy Crossover: After the chromosomes are ranked in terms of their fitness scores, a greedy sequential crossover process is implemented to create the next generation. Similar to that by Al-Furhud [31], an offspring chromosome is generated from two parent chromosomes. Given the first servicing hub, the same first gene is intentionally selected in each chromosome, the next gene selected for the offspring is determined by searching for the best valid gene that follows the previous one in both of the parents. If no valid genes follow the current gene in either of the parents, then the cheapest valid gene is selected from the set of remaining options. While more computationally cumbersome than typical crossover operators, this greedy crossover outperforms other manipulation methods [31] while maintaining valid chromosomes. To select which parent chromosomes create an offspring, the previous generation is ranked in terms of fitness score and are paired off in order of rank, i.e., the first two chromosomes are paired, then chromosomes three and four, and continuing.

As an example, suppose a greedy crossover is implemented on the following two parent chromosomes, $P_1 = [5, 3, 2, 1, 4]$ and $P_2 = [5, 4, 2, 3, 1]$. The child is initially assigned the first gene of 5, with the other genes unassigned, $C = [5, *, *, *, *]$. Then, the cost to transfer from orbit 5 to orbits 3 and 4 are constructed and the minimum solution is selected. Assume that the minimum next orbit is orbit 3, then the child is composed of $C = [5, 3, *, *, *]$, and the process repeats. Then the cost to transfer from orbit 3 to orbits 2 and 1 are compared because they follow the 3 value in both of the parent chromosomes. Assuming, again, that orbit 1 represents the lower cost, the child then becomes $C = [5, 3, 1, *, *]$. At this point, there are no longer valid genes in both parents (since P_2 reached the end of its sequence), so the next gene is selected from the list of remaining legitimate orbits (orbits 2 and 4). This process repeats until the offspring chromosome is filled to the same length as the parents. While this exam-

ple does not include the complications of zero genes representing phasing revolutions and the -1 termination gene, in practice, this process is extended to accommodate such options. When only phasing revolutions and the termination gene are left in the set of legitimate remaining genes, the sequence is terminated and all phasing revolutions, $0s$, are placed after the -1 in the chromosome. While this implementation introduces a bias to reduce the number of phasing revolutions in a schedule over the course of the GA, this bias is only impactful when shorter time-of-flight options also reduce the total ΔV .

Mutation: Finally a mutation operator is also included. The mutation operator in a Genetic Algorithm (GA) is a key mechanism that introduces variability into the population of solutions. After the selection and crossover steps, mutation randomly alters some genes in the chromosomes of the offspring. This process mimics biological evolution where small random changes in genetic material can lead to new traits. The purpose of mutation is the maintenance of genetic diversity within the population, aiding the avoidance of premature convergence to a local optimum by exploring a broader solution space. Mutation is implemented by randomly selecting two genes in the chromosome, except the first value and the termination gene, and swapping their locations in the vector, ensuring that the chromosome remains valid. A potential invalid mutation occurs when a mandatory customer gene is placed behind the termination gene, indicating that it never gets serviced. Mitigating this potential issue, the mutation operator diversifies the search space by swapping servicing orders, increasing or decreasing the number of phasing revolutions, and including or excluding optional customer orbits.

Elitism: Elitism ensures that the best-performing individuals from one generation are preserved and carried over to the next generation without undergoing mutation or crossover. This approach guarantees that the optimal solutions generated thus far are not lost due to the randomness of genetic operations, thus maintaining or improving the overall quality in the population over successive generations. In practice, elitism involves selecting a few top-ranking individuals based on their fitness scores and directly copying them into the new generation. By safeguarding these elite individuals, elitism accelerates convergence towards an optimal solution, as it prevents the degradation of high-quality solutions that might otherwise occur through the GA's stochastic processes. This technique balances the exploration of new solutions with the exploitation of known high-quality solutions, building a more robust and effective algorithm. Elitism is implemented by selecting the best chromosomes from the set of original chromosomes, crossover offspring, and mutations until the desired population is reached. In the event that the three sets do not contain sufficient non-duplicated solutions to reach the required population size, randomly generated solutions are introduced.

Immigration: Finally, immigration is implemented, that is, the introduction of new, randomly generated individuals into the population during the evolutionary process. This technique injects diverse genetic material into the population to prevent premature convergence to local optima and maintain diversity. By occasionally replacing some of the current population with new individuals, the algorithm explores new regions of the solution space not previously examined. Immigration is particularly useful in overcoming stagnation when the population becomes too homogeneous and

with no significant progress toward better solutions. Immigration is implemented in different ways, e.g., replacing a certain percentage of the population at regular intervals or introducing new individuals when the algorithm detects that the population diversity has fallen below a certain threshold. Here, when the maximum score solution remains constant over forty generations, the bottom scoring half of the next generation is replaced with randomly selected chromosomes. By balancing exploration and exploitation, immigration ensures that the GA continues to search for better solutions while avoiding a trap in suboptimal regions of the solution space.

5.3 Comparisons

While the AA provides the general advantage of significantly lower computational costs than the GA, it does not perform as well when designing schedules that involve sequential transfers. The AA is a class of algorithms to solve assignment problems, where the goal is to optimally match a set of agents to a set of tasks, often based on costs or profits associated with each assignment. These algorithms are inspired by auction processes, where the customers field bids by servicers to maximize their own profit or minimize their cost. Since the problem of designing a servicing network is more similar to the TSP, naturally, the GA accommodates downstream dependencies better than the AA. In addition, the previously proposed AA does not add phasing revs or include optional orbits. While the AA is replaced with a GA for determining the servicer schedules, the transfer selection and design process incorporating the AA is replicated here.

To compare the results of servicer schedules determined by the AA and the GA, a servicing scenario is proposed. A network comprised of two servicing hubs and three customer satellites placed in Lyapunov orbits around the L_1 and L_2 Lagrange points is selected for this example. Visualized in Fig. 21, the customer satellites (Blue) are placed in L_1 and L_2 Lyapunov orbits at the JC value $JC = [3.16, 3.14, 3.13]$ and the servicing hubs (Red) are placed in similar orbits at the JC value $JC = 3.15$. After the orbits are selected, the AA and GA are used to design schedules for the two servicers to service all customer spacecraft before returning to their respective hubs. The parameters used for the GA are adjusted to no longer feature exploration of the

Fig. 21 Example Lyapunov network, Moon-centered CR3BP. Red: servicing hub orbits; Blue: customer orbits

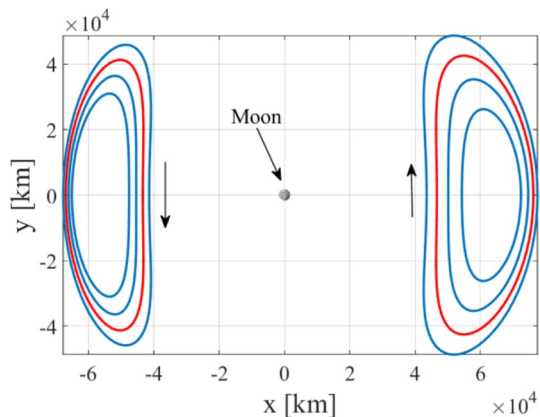


Table 2 GA parameters

Parameter	Value
ΔV Weighting (Eq. 9)	0.15
γ	0
γ_{JC}	0
γ_{SK}	0
γ_{SP}	0
Generations	500
Population	30
Immigration rate	40 generations
Immigration replacement rate	60 %

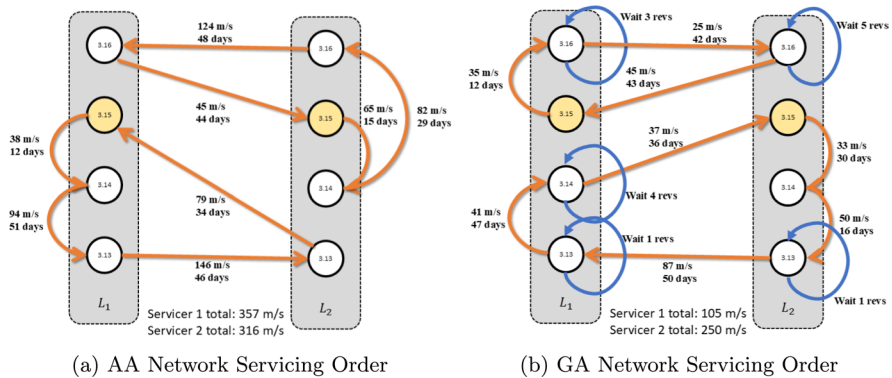


Fig. 22 Illustrative schedules produced by the AA (a) and the GA (b). Initial servicing hub orbits in orange and customer orbits in white; one each near L_1 and L_2 . Orange arrows indicate transfers taken by servicer satellites. Phasing revolutions are indicated in blue

orbit space, but exploit low-cost transfer instead; the parameters are summarized in Table 2.

Upon conclusion, the AA and GA produce two different networks in terms of the order in which the customers are serviced and the different assignments for the servicers. A simplified schematic of the servicing order selected by both methods appears in Fig. 22 with the AA schedule on the left and the GA schedule on the right. In the figure, the ΔV is indicated for the fully converged network in the CR3BP.

As is noticeable by examining the servicer ΔV totals at the bottom of each sub-figure, the GA network requires significantly less ΔV , but also requires more time to service the entire network. The fitness scores for each generation of the GA along with the score associated with the AA are displayed in Fig. 23. It is clear from the figure that, after the first generation of the GA, the average score significantly outperforms that of the AA. In fact, even the best scoring chromosome of the first generation already outperforms the AA.

While the GA was continued for all 500 generations, it is clear that the average value starts to plateau around the 150th generation, indicating that, although the maximum score slightly increases afterward, the GA could be stopped at this point for computational efficiency. Also apparent are the eight sharp downward spikes in the average scores across the generations due to the implementation of immigration

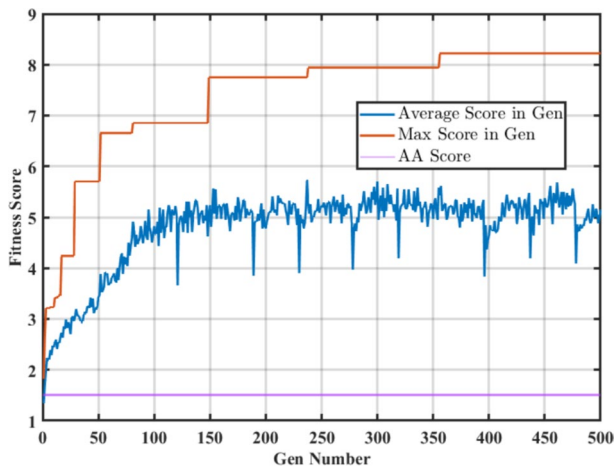


Fig. 23 Evolution of GA fitness score and score of AA network. The panels highlight the impact of using (i) a fast allocator vs. (ii) a tour optimizer under time-dependent, phasing-aware edges. Results are shown for context, not as a head-to-head benchmark. Blue: Average generation fitness score. Orange: Maximum generation fitness score. Purple: Fitness score of AA schedule

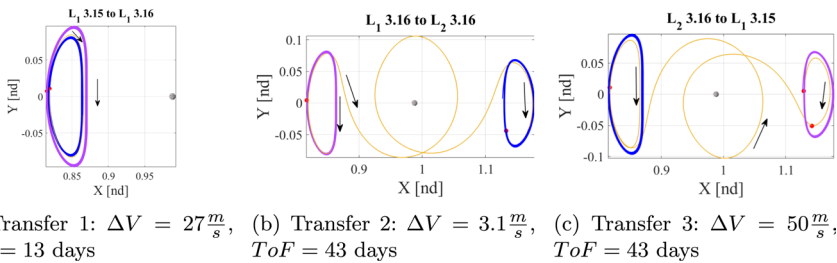


Fig. 24 Ephemeris transfers for servicer 1 starting in L_1 Lyapunov at JC = 3.15 visualized in the pulsating-rotating frame [32]. Servicing order proceeds from left to right. Purple: initial orbit, Blue: target orbit, Yellow: transfer trajectory, Red Dots: ΔV locations

after 40 generations for the same maximum solution. In this example, the injection of random individuals sometimes leads to the crossover result for a new maximum score, but not in all cases. After taking the previously converged GA solution and transitioning it to the ephemeris model, the required ΔV is further reduced. The final ephemeris solution for servicer 1 is plotted in Fig. 24 and servicer 2 in Fig. 25. The total ΔV and transfer time for each servicer is included in Table 3. In the table, comparisons between the CR3BP and ephemeris values are provided.

As noted in Waldecker and Howell [19], the ephemeris transition process involves translation to the CR3BP barycenter and rotation to match the desired epoch, for this network, September 15, 2025. The underlying orbits are converged for a one year baseline solution and the transfers are constrained to apply a departure and arrival burn along the baseline orbits such that the departure and arrival locations match that of the previous and upcoming spacecraft respectively.

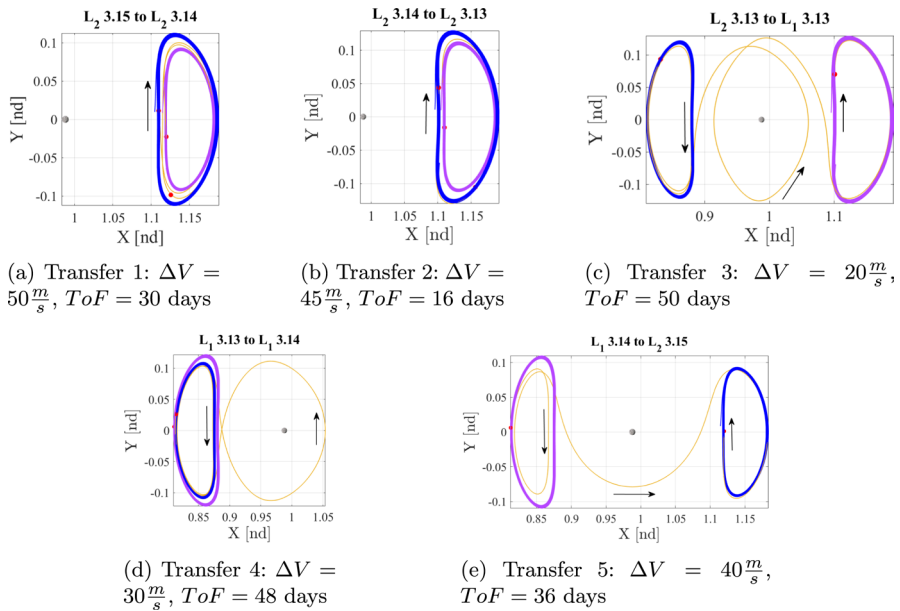


Fig. 25 Ephemeris transfers for servicer 2 starting in L_2 Lyapunov at $JC = 3.15$ visualized in the pulsating-rotating frame [32]. Servicing order proceeds from left to right. Purple: initial orbit, Blue: target orbit, Yellow: transfer trajectory, Red Dots: ΔV locations

Table 3 Transfer metrics

	CR3BP $\Delta V [\frac{m}{s}]$	CR3BP ToF [d.]	Ephem. $\Delta V [\frac{m}{s}]$	Ephem. ToF [d.]
<i>Servicer 1</i>				
L_1 3.15 to L_1 3.16	35.9	12.156	27.4	12.729
L_1 3.16 to L_2 3.16	25.4	42.356	3.2	43.159
L_2 3.16 to L_1 3.15	44.6	42.554	49.9	43.465
<i>Servicer 2</i>				
L_2 3.15 to L_2 3.14	33.2	29.912	50	30.100
L_2 3.14 to L_2 3.13	50.0	15.549	45.2	15.648
L_2 3.13 to L_1 3.13	87.0	47.329	20	50.476
L_1 3.13 to L_1 3.14	41.9	47.329	29.7	48.132
L_1 3.14 to L_2 3.15	37.9	35.616	40	35.625

6 Results and Discussion

The application of the GA across various servicing networks is also explored, demonstrating its robustness and adaptability to different types of orbital configurations and the ability to select initial servicer/customer orbits that maximize network benefit. The flexibility of the GA in accommodating diverse orbital configurations is a key element for optimizing satellite servicing networks, especially in the complex gravitational environment near the Moon. The GA consistently demonstrates its efficiency in minimizing the overall ΔV across multiple scenarios with distinct orbital charac-

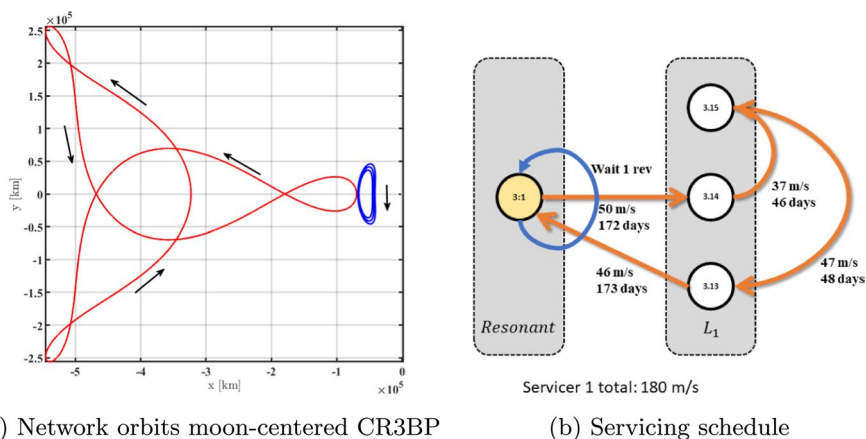


Fig. 26 **a** Network with servicing hub in 3:1 resonant orbit and three customer satellites in L_1 Lyapunov orbits and **b** servicing schedule. Servicer hub in yellow; customer orbits in white

Table 4 3:1 Network Transfer Metrics

Servicer 1	CR3BP $\Delta V [\frac{m}{s}]$	CR3BP ToF [d.]	Ephem. $\Delta V [\frac{m}{s}]$	Ephem. ToF [d.]
3:1 to 3.14	50	172.734	80.6	172.69
3.14 to 3.15	37.1	45.9	35.5	46.16
3.15 to 3.13	46.9	47.698	30	48.08
3.13 to 3:1	46.1	173.554	55.3	173.97

teristics. This adaptability highlights the GA's effectiveness in optimizing servicing schedules and its potential to be applied to a wide range of architectures.

6.1 Resonant Orbits and Lyapunov Orbits

The first additional network involves the placement of a servicer hub in a 3:1 resonant orbit with a $JC = 3.15$ and customer satellites placed in L_1 Lyapunov orbits at $JC = [3.14, 3.15, 3.16]$. Resonant orbits present particular challenges when they are incorporated into a servicing network due to the inherently long periods. While they offer numerous transfer options over the span of a single period, the resulting transfer times are frequently much longer than other orbits in the lunar vicinity. Representative orbits are plotted in the CR3BP in Fig. 26(a) and the network determined by the GA is displayed in Fig. 26(b). Though the resonant orbit does not offer the most favorable phasing characteristics, the GA is capable of identifying a network for less than $200 \frac{m}{s}$ total, leveraging only one phasing revolution at the start of the mission scenario.

After converging the network in the ephemeris model, it is noticeable in Table 4 that the propulsive costs departing from and arriving to the resonant orbit increase. This ΔV increase is likely due to the difficulty in converging the two-year baseline for the resonant orbit, creating large position discontinuities at the initiation of the convergence process. In addition, due to the vicinity of the 3:1 resonant orbit,

an Earth-centered propagation and targeting scheme is exploited as opposed to the previously used Moon-centered scheme. The ephemeris transfers throughout the network appear in Fig. 27.

6.2 Network Variants

In addition to planar and resonant orbits, the methodology is promising for the design of spatial servicing networks as well. A network comprised of one servicing hub in a butterfly orbit and another in a P2HO2 at $JC = 3.05$ serves as such an example. Three customers are placed in the same three halo orbits used for the auction algorithm, i.e., L_1 halo, L_2 halo, and L_2 NRHO halo orbits. From the results of the scheduling algorithm, the butterfly servicer first transfers to the L_1 halo orbit customer, then to the L_2 NRHO before returning to the servicing hub in the butterfly orbit, as is displayed in Fig. 28. The second servicer starts in the P2HO2, transfers to the L_2

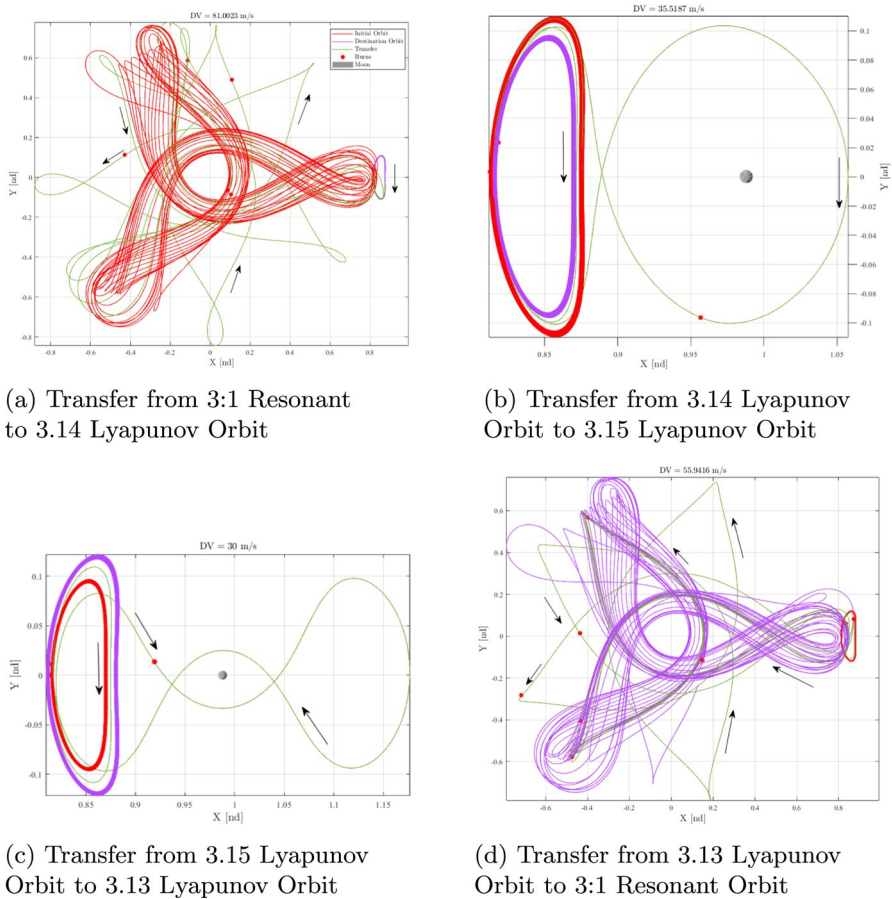
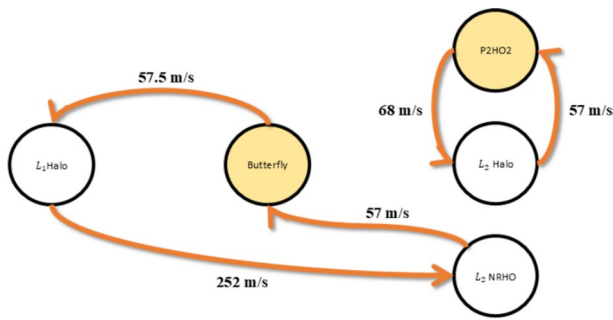


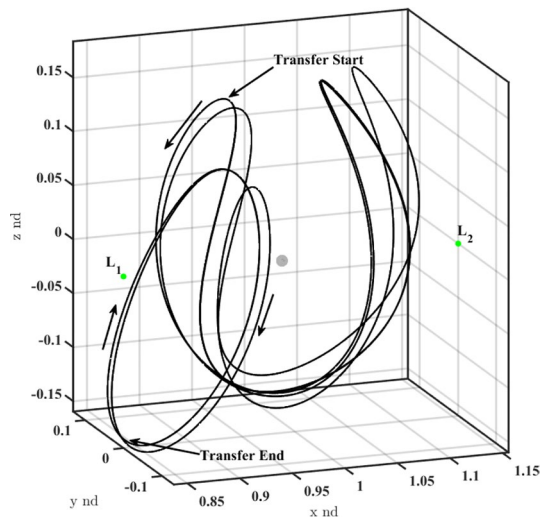
Fig. 27 Ephemeris converged transfers throughout the 3:1 network plotted in the pulsating-rotating frame. In each sub figure: Red - Initial Orbit, Purple - Target Orbit, Green - Transfer Trajectory, Red Dots - ΔV Locations



Servicer 1 total: 366.5m/s
Servicer 2 total: 125 m/s

Fig. 28 Spatial AA Network. Servicer hubs in yellow; customer orbits in white

Fig. 29 Servicer 1 transfer 1 to L_1 halo CR3BP. ToF = 85 days, $\Delta V = 57.5 \frac{m}{s}$



halo orbit, and returns to the original hub. For simplicity, it is assumed that the servicing occurs over the course of one periodic revolution for each customer orbit and no phasing revolutions are included in the process.

6.2.1 Servicer 1

Servicer 1 originates in the butterfly orbit and initially transfers to the L_1 halo orbit customer, as plotted in Fig. 29. Notably, and not surprisingly, this result is one of the same transfers demonstrated in Fig. 13. After completing the servicing over the course of one revolution, roughly 12 days, Servicer 1 then transfers to the L_2 NRHO as viewed in Fig. 30. As designed, the total time to transfer to the L_1 halo orbit, one

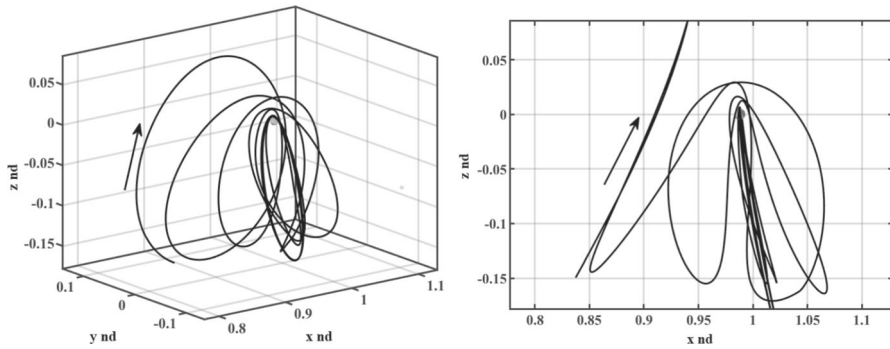


Fig. 30 Servicer 1 transfer 2 to L_2 NRHO CR3BP. ToF = 62.35 days, $\Delta V = 252 \frac{m}{s}$

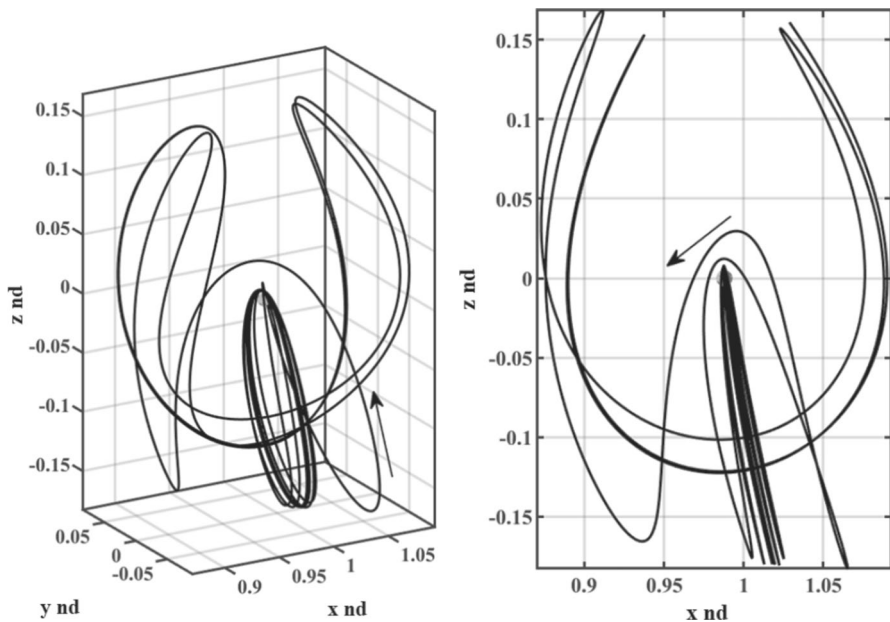


Fig. 31 Servicer 1 transfer 3 to butterfly orbit CR3BP. ToF = 104 days, $\Delta V = 57 \frac{m}{s}$

period along the L_1 halo orbit, and the time to transfer to the L_2 NRHO is exactly 25 periods (about 159 days) of the NRHO, indicating proper phasing for rendezvous.

After accomplishing the servicing for the NRHO customer, Servicer 1 returns to the butterfly orbit and completes a rendezvous in the original servicing hub. The return transfer is plotted in Fig. 31. The minimum altitude along both the arrival and departure trajectories from the NRHO are nearly 300 km from the Moon's surface.

The complete transfer time for Servicer 1 to reach and service both assigned customers and return to the servicing hub orbit is 251 days and requires a total $\Delta V = 367 \frac{m}{s}$, propellant that must be stored in the single servicer spacecraft before being refueled at the servicing hub.

6.2.2 Servicer 2

As Servicer 1 initiates its assignment, Servicer 2 simultaneously departs the P2HO2 and services one customer (in the L_2 halo orbit) before returning to the original orbit for servicer 2. Due to the similarity of the two orbits, the transfer costs in the CR3BP are relatively low. The first transfer from the P2HO2 to the customer in the L_2 halo orbit is plotted in Fig. 32.

Following a servicing time of one period in the L_2 halo orbit, the servicer returns to the servicing hub as observed in Fig. 33.

The total ΔV required for Servicer 2 is $125 \frac{m}{s}$ and the total transfer time throughout the network is 77.4 days. As expected, the two transfers are nearly symmetrical due to the phasing properties of the two orbits.

In combination, Servicer 1 and Servicer 2 offer a servicing network capable of servicing customers in L_1 halo orbits, L_2 halo orbits and L_2 NRHOs. Each servicer satellite individually requires less than $370 \frac{m}{s}$ to transfer to each customer with the highest cost customer located in the NRHO. The large cost is not due to the energy change, since all the orbits have the same JC values, but primarily due to the large inclination changes required to reach the NRHO. In total, the two servicers are capable of providing services to every customer in the network in under one year.

6.3 South Pole Optimized Network

Additionally, a network of servicer hubs placed in a northern butterfly orbit at $JC = 3.05$ and an L_1 Lyapunov orbit at $JC = 3.06$ with customer satellites in: a northern butterfly orbit at $JC = 3.04$, a southern L_2 halo orbit at $JC = 3.05$, an L_1 Lyapunov orbit at $JC = 3.05$, and an L_2 Lyapunov orbit at $JC = 3.04$, are evaluated to demonstrate the versatility of the GA. This network of orbits is selected as an

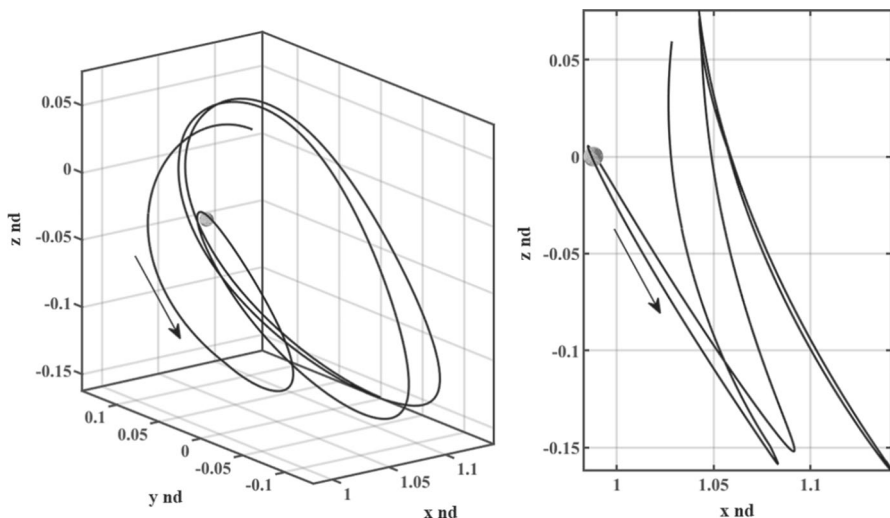


Fig. 32 Servicer 2 transfer 1 to L_2 halo CR3BP. ToF = 39 days, $\Delta V = 68 \frac{m}{s}$

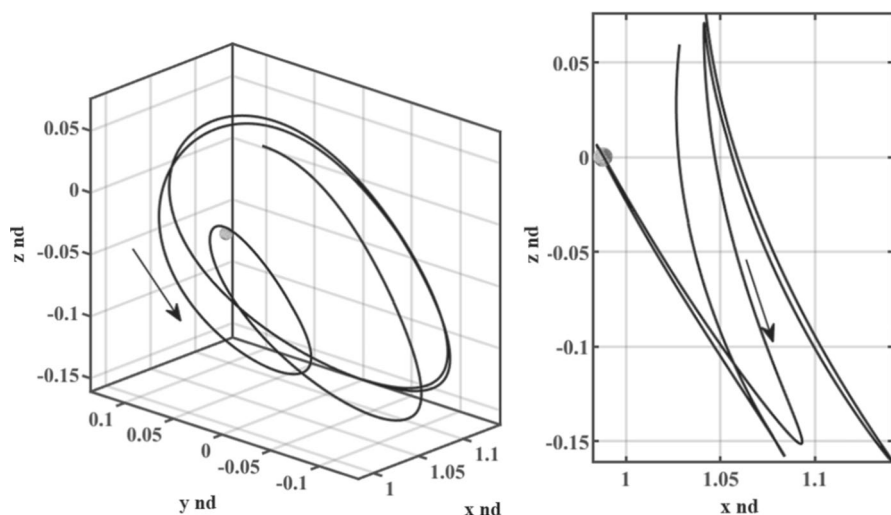


Fig. 33 Servicer 2 transfer 2 to P2HO2 CR3BP. ToF = 39 days, $\Delta V = 57 \frac{m}{s}$.

Table 5 GA parameters

Parameter	Value
ΔV Weighting (Eq. 9)	0.15
γ	1
γ_{JC}	1
γ_{SK}	1
γ_{SP}	20
Generations	500
Population	30
Immigration rate	40 generations
Immigration replacement rate	60 %

optimal network that heavily weights Lunar south pole (SP) coverage. The parameters for this network are displayed in Table 5.

In the table, it is noticeable that the weighting for Lunar south pole coverage is significantly larger than the other values. This large value intentionally identifies the specific orbits that offer the most coverage, while also requiring the least amount of ΔV . Through a process of seeding various orbits in the lunar vicinity, the final network is selected emerging from 100 random seeds. Due to the nature of low ΔV transfers between halo orbits near the same Lagrange point and the energy level, the search space is pruned to remove networks solely comprised of similar halo orbits. The final network is displayed in Fig. 34 along with the converged ΔV requirements.

Due to the heavy weighting of the Lunar south pole coverage, the ΔV requirements to transfer between the spatial orbits is significant. However, due to the large time interval that the halo orbits remain in view and, thus, maintain coverage of the Lunar south pole, the algorithm identified the additional transfer as a part of the optimal configuration. Against natural intuition, the GA also converged to a solution that included a servicer and customer satellites placed in Lyapunov orbits, even though

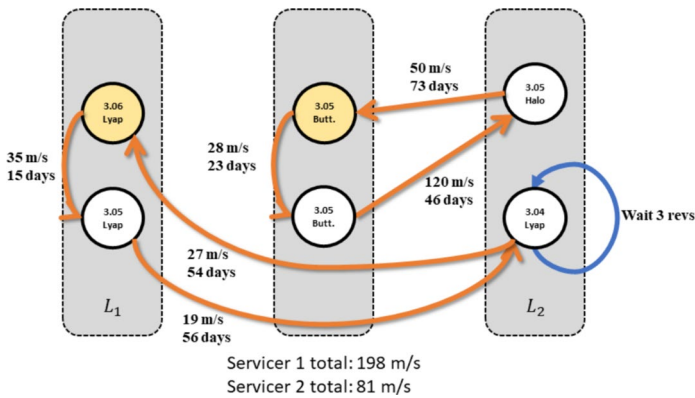


Fig. 34 Lunar south pole coverage optimized network. Recall that servicers are yellow; customers in white

they offer no Lunar south pole coverage. While the orbits offer no benefit in terms of south pole coverage, the significantly lower ΔV required, low deployment costs, and low station keeping costs makes these orbits suitable candidates as part of the servicing network. Finally, while the mission objective for this example is to maximize Lunar south pole coverage, the GA fitness score may be altered to meet the needs of the designer.

6.4 Ephemeris Validation

To validate the results from the CR3BP model, the orbits and transfers from Fig. 29 to Fig. 33 are transitioned into an Earth-Moon-Sun Ephemeris model with the initial epoch equal to September 15, 2025. The process for generating the baseline solutions involves selecting a desired initial epoch (usually a date that aligns with the Earth-Moon distance, l^* , in the CR3BP), stacking revolutions of the desired orbit, and converging using a multiple-shooting corrections algorithm for full state ballistic continuity [33–35]. Other investigations focus on pre-determined epoch dates; such details are beyond the scope of this current analysis. The one year baseline trajectories, as constructed in the ephemeris model, appear in Fig. 35.

Then, each of the original transfers is transitioned into the ephemeris model and converged. To constrain the transfers between two fixed orbits, the initial and final state along the transfer must be fixed. However, fixing the end nodes of a transfer constrains the problem significantly and frequently results in excessively high ΔV solutions. This constraint is overcome by fixing the underlying orbits, allowing the initial and final states to slide along the respective orbits, and varying the transfer time to achieve rendezvous. Since the transfers in the CR3BP are defined from apolune to apolune, the epoch, appropriate for the apolune states along the departure and arrival orbits, are leveraged as the initial guess for the end nodes of the transfer. Note that none of the ephemeris network solutions are optimized. The ΔV is occasionally reduced using slack variables, however, some of the higher cost solutions are likely to be improved via an optimization scheme. As a sample transfer to highlight the results,

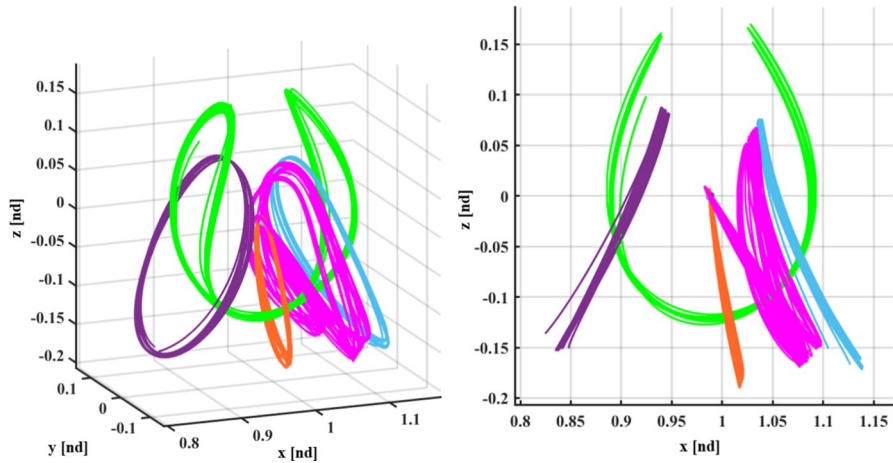


Fig. 35 One year ephemeris converged baseline orbits CR3BP. Green: butterfly orbit, Purple: L_1 halo orbit, Orange: L_2 NRHO, Magenta: P2HO2, Blue: L_2 halo orbit

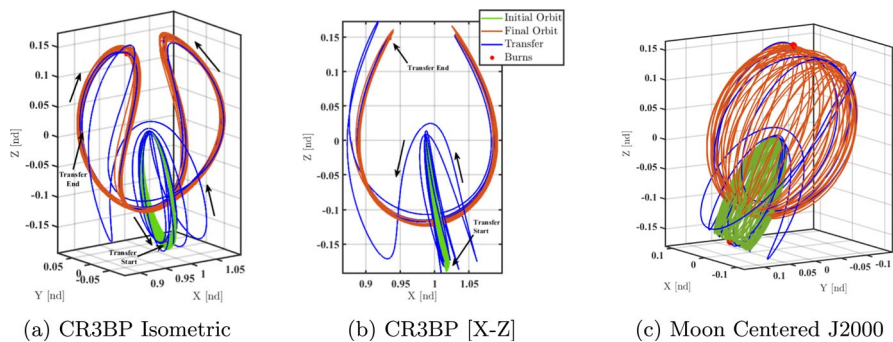


Fig. 36 E-M-S transfer from L_2 NRHO to butterfly orbit. Brown: butterfly orbit, Green: L_2 NRHO, Blue: converged transfer, Red: ΔV locations

the return path from the L_2 NRHO to the butterfly servicing orbit is plotted in Fig. 36, illustrating strong similarity to the geometry from the CR3BP. Comparisons between transfers in the CR3BP and the ephemeris counterparts are tabulated in Table 6.

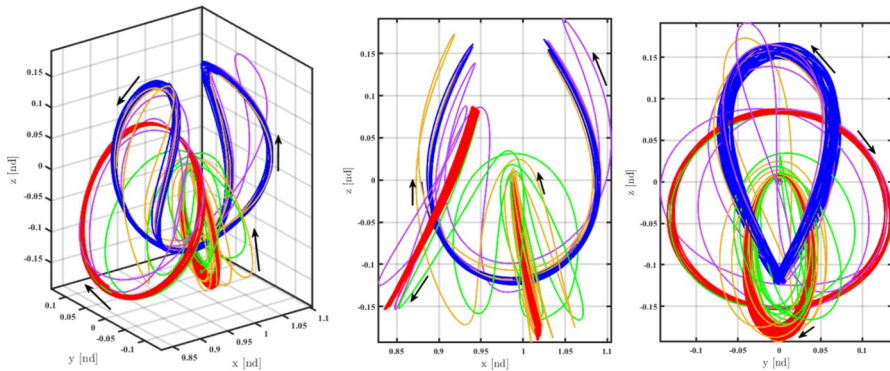
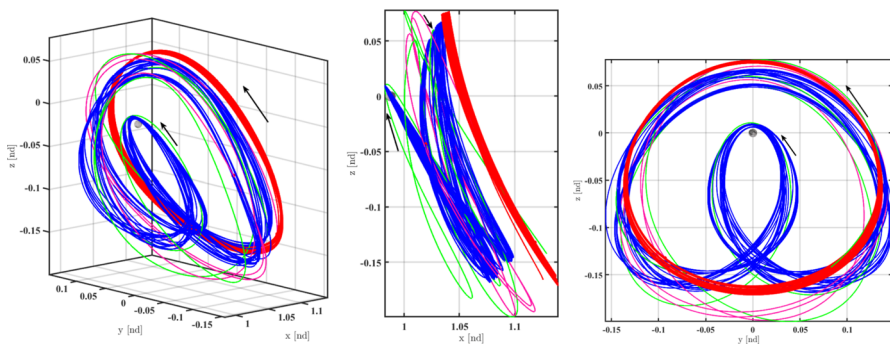
The original transfer in the CR3BP required 104 days to complete and $57 \frac{m}{s}$ of ΔV . After convergence in the ephemeris model, the transfer requires 105 days and $\Delta V = 50 \frac{m}{s}$; the applicability of using the CR3BP for initial modeling is apparent.

6.4.1 Servicer 1

Converging the same transfer network for Servicer 1 in an E-M-S Ephemeris model (Fig. 37), many of the geometries and attributes are similar. The required ΔV for each transfer is $69 \frac{m}{s}$, $220 \frac{m}{s}$, and $50 \frac{m}{s}$, a reduction of $28 \frac{m}{s}$ over the CR3BP counterparts while requiring 255 days to complete, a slight increase.

Table 6 CR3BP and ephemeris transfer characteristics

Transfer	CR3BP ΔV	Ephem. ΔV	CR3BP ToF	Ephem. ToF
butterfly to L_1 halo	$57.5 \frac{m}{s}$	$69 \frac{m}{s}$	85 days	86 days
L_1 halo to L_2 NRHO	$252 \frac{m}{s}$	$220 \frac{m}{s}$	62 days	63 days
L_2 NRHO to butterfly	$57 \frac{m}{s}$	$50 \frac{m}{s}$	104 days	105 days
P2HO2 to L_2 halo	$68 \frac{m}{s}$	$145 \frac{m}{s}$	39 days	40 days
L_2 halo to P2HO2	$57 \frac{m}{s}$	$180 \frac{m}{s}$	39 days	38 days

**Fig. 37** E-M-S servicer 1 network and transfers visualized in the CR3BP. Blue: servicing orbit, Red: customers orbits, Purple: butterfly orbit - L_1 halo orbit transfer, Green: L_1 halo orbit - L_2 NRHO transfer, Gold: L_2 NRHO - butterfly orbit transfer**Fig. 38** E-M-S servicer 2 network and Transfers Visualized in the CR3BP. Blue: Servicing Orbit, Red: Customer Orbit, Green: P2HO2 - L_2 halo orbit transfer, Pink: L_2 halo orbit - P2HO2 transfer

6.4.2 Servicer 2

In contrast to the Servicer 1 network, the propulsive requirement increased when converging the Servicer 2 network in the ephemeris model (Fig. 38). The transfers between the P2HO2 and the L_2 halo orbit increased from $68 \frac{m}{s}$ and $57 \frac{m}{s}$ to $145 \frac{m}{s}$ and $180 \frac{m}{s}$ respectively, nearly tripling the total cost of the Servicer 2 network. The

Servicer 2 network maintains nearly the same transfer time of 78 days. Adjusting the departure time of the servicer from the hub may possibly mitigate such an increase.

7 Concluding Remarks

This investigation introduces a comprehensive framework for on-orbit servicing network design, utilizing genetic algorithms (GA) for orbit selection, phasing-informed transfer design strategies, and robust scheduling methods. By adhering to beneficial criteria for network design, the GA reduces the orbit search space and enables the identification of high-performing solutions that belong to similar families and exhibit comparable energy levels. The GA fitness evaluation incorporates deployment costs, station-keeping requirements, and Lunar south pole coverage, while remaining extensible to other mission objectives.

The proposed transfer design pipeline leverages manifold arc trajectories to generate heatmaps for transfer options. By incorporating a phasing penalty, it efficiently estimates the transfer costs for long-range rendezvous without requiring convergence of hundreds of transfer options. This approach offers greater accuracy in estimating the true cost of converged transfers under phasing constraints and enhances intuition of the underlying dynamics when designing network schedules.

For scheduling, two solutions are presented: a low-compute auction algorithm (AA) and a higher-computation GA. These strategies enable the design of servicing schedules that maintain dynamic intuition and are more informed by the dynamics in the environment. The scalability of these schemes supports diverse orbital configurations and varying numbers of satellites, facilitating trade space analyses to identify orbit configurations that meet mission-specific requirements. A direct GA over ephemeris dynamics is conceptually feasible but presently impractical for the network scales considered: each cost evaluation entails high-order N -body integration and multi-point shooting with variational states, and robust convergence depends on CR3BP-quality initial guesses. The two-stage pipeline presented reduces search cost while preserving fidelity.

When ground computational capabilities are available, the unique genetic algorithm implementation offers promising results. The GA employs a one-dimensional chromosome representation to encode servicer orbits, customer orbits, phasing revolutions, and optional orbits for trade space analysis. With selection, crossover operators, and elitism, the GA determines effective servicer satellite schedules for diverse network configurations. Applied to L_1 and L_2 Lyapunov orbits, the GA significantly outperforms an auction algorithm by incorporating downrange phasing requirements. In the absence of standardized benchmarks for cislunar servicing networks, order-of-magnitude runtime guidance and the principal scaling drivers rather are provided. The pipeline's cost is dominated by (i) the number of candidate transfers constructed by the phasing-informed generator and converged via CR3BP differential corrections, and (ii) the size of the dynamic graph explored by the scheduler (servicer/customer counts) together with GA settings (population, generations, and early-stopping criteria). On the machine used (Windows, Intel i7-6600U, 8.00 GB), auction-based seeding typically completes on the order of one minute, and GA tour optimization is

on the order of tens of minutes for the networks reported herein. Because absolute times vary with hardware and problem size, scaling is emphasized rather than fixed numbers.

The GA leverages a multi-objective fitness score, that incorporates total network ΔV , servicer count, and customer orbit benefit. This evaluation emphasizes minimizing deployment costs and orbital instability while maximizing coverage of the Lunar south pole. Analysis reveals that orbit symmetry (e.g., orbits from the same family) and the difference in energy (i.e., Jacobi Constant gap) renders the largest impact on ΔV requirements, while factors such as time constants and the periods of linearly unstable orbits exhibit negligible influence on transfer costs.

To demonstrate the robustness of the methodology, multiple servicing networks are introduced, including a two-servicer, three-customer network using halo orbits and a P2HO2. This network highlights a low-cost servicing strategy, requiring less than $350 \frac{m}{s}$ of ΔV annually for halo orbits. Heat maps as an approximation for transfer costs between network orbits provide a foundation for initial solutions, that are refined through the GA to minimize ΔV while addressing phasing challenges.

The final network solution is demonstrated in an Earth-Moon-Sun ephemeris model, producing a one-year servicing schedule. Overall, the proposed scheme demonstrates robustness and adaptability across planar, spatial, and resonant orbits, offering a flexible approach to optimize servicing networks for mission-specific objectives, such as Lunar south pole coverage. This framework demonstrates a versatile and robust methodology for designing, optimizing, and evaluating on-orbit servicing networks, providing critical insights into orbit selection, trajectory design, and servicer scheduling to achieve diverse mission objectives.

Beyond the technical optimization of orbits, transfers, and schedules, the viability of cislunar servicing architectures also hinges on a resilient logistics and supply-chain backbone that can sustain the servicer fleet over multiple years. Future constellations must incorporate propellant and spare-part depots; launch-cadence flexibility, including ride share and dedicated small-launch options, to replenish consumables; common-bus standards and modular interfaces that allow servicers built by different vendors—or produced in staggered batches—to dock interchangeably with customer spacecraft; and diversified supply chains for critical items. The future of on-orbit servicing hinges on the culmination of technological advancements across various space and terrestrial fields, but offers the potential to transform satellite operations.

Author Contributions C.W. and K.H. contributed equally to this work.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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